Sectoral Labor Reallocation, Inflation and Monetary Policy*

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Abstract

We study how frictional labor reallocation shapes the inflation dynamics and monetary policy using a multi-sector New Keynesian economy. We derive analytical expressions for the Phillips curve and welfare, demonstrating that the frictional labor mobility influences sectoral inflation dynamics through two sufficient statistics: a mobility elasticity and the steady-state labor transition matrix. Using a spectral decomposition, we show that the persistence of sectoral shocks is determined by the interaction between frictions in labor mobility and nominal rigidity. When labor reallocation across sticky sectors is costly, the effect of sectoral shocks become more persistent. In response to sectoral shocks, monetary policy balances stabilization of standard aggregate output gaps with persistent labor and price misallocation. When these sticky sectors face a negative productivity shock, optimal monetary policy has an extra incentive to tighten to decrease the relative demand and thereby incentivize labor outflows.

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1 Introduction

Sectoral shocks are pervasive: energy prices spike, supply chains break, demand rotates across goods and services, and technologies reshape which tasks are valuable. These events call for relative adjustment—prices realign and workers move toward expanding activities. In a frictionless benchmark, both margins operate efficiently. In practice, both margins are sluggish. On the nominal side, sticky prices delay relative price adjustments, generating distortions that misallocate expenditure. On the real side, worker reallocation unfolds gradually through a complex network of bilateral flows shaped by skills, geography, and institutional barriers.

These two frictions are individually well established. Price stickiness is the foundation of the New Keynesian framework, and labor reallocation frictions are foundational in the trade and labor literature. In reality, the two margins interact: price distortions contaminate the signals that guide reallocation, while labor frictions feed back into the cost pressures that drive inflation. Analyzing these margins in isolation misses a feedback loop that governs the economy's ability to absorb sectoral shocks.

When sectors differ in price rigidity, aggregate demand affects prices unevenly. As a result, monetary policy does more than influence the aggregate price level: it changes the relative prices that guide labor reallocation. Policies that focus on aggregate stabilization can impede the relative price movements needed to direct workers toward expanding sectors, while policies that facilitate real reallocation can generate larger inflation differentials across sectors. Which force dominates depends on how sectoral price rigidities map onto the labor flow network, and the interaction of nominal rigidities and labor reallocation frictions shapes the transmission of sectoral shocks, the persistence of inflation, and optimal monetary policy.

We study these interactions in a multi-sector New Keynesian model with frictional labor reallocation and establish three main results. First, relative to a benchmark with fully integrated labor markets, segmented labor markets amplify the initial inflation response to sectoral shocks but speed up subsequent convergence. Second, the persistence of relative inflation depends neither on average price rigidity nor on aggregate labor mobility, but on the price rigidity of the sectors that form reallocation bottlenecks. When workers must reallocate through a corridor connecting two sticky-price sectors, both frictions compound and adjustment stalls. Third, heterogeneity in sectoral price rigidity causes optimal monetary policy to deviate from the textbook targeting rule. The planner tightens when efficient reallocation requires workers to leave sticky-price sectors and loosens when they must leave flexible-price sectors, thereby exploiting the unequal pass-through of aggregate demand into relative prices.

In Section 2, we build a multi-sector New Keynesian model that incorporates frictional labor reallocation through an Artuç et al. (2010) dynamic discrete choice block. Workers choose sectors based on forward-looking continuation values, subject to bilateral mobility costs and idiosyncratic preference shocks. Despite the richness of the underlying micro structure, the

linearized equilibrium and the policy problem depend on just two sufficient statistics: a mobility elasticity ϕ that governs the responsiveness of aggregate flows to sectoral value differentials, and the steady-state labor transition matrix $\bar{\mu}$ that summarizes the pattern of gross worker flows across sector pairs.

Section 3 analyzes a static benchmark with within-period labor reallocation. In this limit, the effect of mobility frictions is summarized by a single sufficient statistic, the mobility elasticity, which enters both the sectoral Phillips curve and the welfare loss function. The mobility elasticity determines how adjustment splits between worker flows (the extensive margin) and hours (the intensive margin). When mobility is low, adjustment falls on the intensive margin, raising the welfare cost of relative price distortions. Optimal policy therefore places greater weight on sticky-price sector inflation to limit expenditure misallocation. When mobility is high, worker flows absorb relative demand shifts, reducing the sensitivity of sectoral marginal costs to demand. This steepens the Phillips curve in flexible-price sectors and shifts the optimal targeting weight toward a broader inflation index.

Section 4 extends the analysis to the dynamic economy, where sectoral employment shares become state variables. Under homogeneous price stickiness, aggregate demand shifts marginal costs uniformly across sectors, so the system decouples into an aggregate block that monetary policy controls and a relative block that it cannot influence. A spectral decomposition of $\bar{\mu}$ decomposes the relative block into independent components, each capturing a self-contained pattern of worker reallocation across sectors. At the two polar cases, segmented and fully integrated labor markets, impact and persistence move in opposite directions: segmented markets produce a larger initial inflation response but faster subsequent convergence, because firms that cannot anticipate cost relief from worker inflows reprice more aggressively today. For arbitrary mobility, the persistence of relative inflation within each component is determined by the interaction between the Phillips curve slope κ and the component's labor-mixing eigenvalue ρ_k, with the slower-adjusting margin acting as the bottleneck.

Heterogeneous price stickiness breaks this spectral separation, coupling previously orthogonal components. The first consequence is that the persistence of a sectoral shock is determined by the stickiness of the sectors along the specific path workers must take to reallocate. We formalize this via the *effective rigidity* $\Delta\kappa_k$ of each labor reallocation corridor k. When the sectors along the slowest labor reallocation path also tend to be sticky, adjustment stalls on both margins: sluggish price responses mute the value differentials that induce reallocation, while slow labor flows sustain the marginal cost gaps that drive inflation. We derive this result as a local perturbation; Section 5 evaluates whether it holds under the large dispersion of price stickiness observed in the data.

The second consequence is that aggregate demand shifts relative prices unequally across sectors. The *spillover coefficient* d_k measures whether sticky and flexible sectors sit on opposite sides of a corridor, so that aggregate demand tilts relative prices along that component. Persistence

and spillover can move independently: a corridor connecting two sticky sectors may exhibit high persistence (large negative $\Delta\kappa_k^{\text{eff}}$) but near zero spillover d_k, leaving the central bank unable to steer reallocation along it. Both sufficient statistics are computable from the worker transition matrix and sectoral price adjustment frequencies.

We formalize the optimal policy response as a dynamic Ramsey problem. The second-order welfare loss penalizes cross-sector employment misallocation and relative price distortions—costs that compound over time when labor reallocation is slow. The resulting targeting criterion nests the textbook rule when stickiness is uniform but introduces a reallocation wedge when stickiness is heterogeneous. The planner deviates from the textbook rule when effective rigidity and spillover are both large along the same corridor. Effective rigidity $\Delta\kappa_k^{\text{eff}}$ determines the necessity of intervention: sticky corridors create persistent welfare costs. The spillover d_k determines the direction: when the sticky side of a bottleneck corridor needs to shed workers, the planner tightens to lower relative demand for that sector's output; when workers must instead flow toward the sticky side, the planner loosens.

Section 5 brings the model to U.S. sectoral data, estimating the transition matrix $\bar{\mu}$ from worker flows and measuring the alignment between price rigidity and the network's slowest-mixing corridors to assess the quantitative relevance of these channels.

Related Literature

Our paper is most closely related to work on monetary policy during structural reallocation. Guerrieri et al. (2021) show in a two-sector model with downward wage rigidity that sectoral reallocation can shift optimal policy in either direction: contractionary when easing slows reallocation, expansionary when easing accelerates it. Ferrante et al. (2023) study the interaction of inflation and labor reallocation in a calibrated 66-sector model, showing that labor reallocation frictions combined with heterogeneous price stickiness account for a large share of post-pandemic U.S. inflation. We generalize the policy insight of the former and provide analytical foundations for the mechanism in the latter, deriving sufficient statistics—computable from worker flow data and sectoral price adjustment frequencies—that determine both when reallocation frictions amplify inflation persistence and when they affect optimal policy.

We also contribute to the study of optimal monetary policy in multi-sector New Keynesian economies. Existing work shows how heterogeneous price rigidity (Aoki, 2001) and production networks (La'O and Tahbaz-Salehi, 2022; Rubbo, 2023; Afrouzi and Bhattarai, 2025) shape optimal policy and inflation persistence. These models typically assume labor is either perfectly mobile or permanently segmented across sectors. We show that intermediate, frictional mobility generates distinct channels for both policy and persistence that are absent at either polar case.

Our micro-foundation for labor reallocation builds on the trade and labor literature on dynamic worker reallocation, originating from Lilien (1982) and formalized via dynamic discrete

choice by Artuç et al. (2010) with general equilibrium applications in Caliendo et al. (2019); Allen et al. (2020); Seo and Oh (2025); Kim et al. (2026). We embed this micro-foundation into a New Keynesian model and show that the reallocation block enters the Phillips curve and welfare loss through two sufficient statistics. Methodologically, we adapt spectral decomposition techniques used in spatial migration (Kleinman et al., 2023) and input-output production networks (Liu and Tsyvinski, 2024) to characterize the dynamics of cross-sector worker flows.

We also add to the literature studying how heterogeneous price rigidity (Bils and Klenow, 2004; Nakamura and Steinsson, 2008) and sectoral linkages (Pasten et al., 2020; Pastén et al., 2024) shape macro dynamics. Prior work shows that heterogeneous propagation causes inflation persistence (Altissimo et al., 2009) and that labor market segmentation matters for sectoral adjustment (Carvalho et al., 2021). We show that the topology of worker flows—which sectors are connected to which, and how quickly workers move between them—interacts with price stickiness to create a distinct source of relative inflation persistence.

2 Model

We build a multi-sector New Keynesian model in which sectors differ in price rigidity and workers reallocate across sectors through a dynamic discrete choice problem. The model combines standard Calvo pricing on the nominal side with an Artuç et al. (2010) reallocation block on the real side. We describe the production side, household preferences, and the labor reallocation problem, then state the equilibrium and its log-linear approximation.

Time is discrete, $t = 0, 1, \ldots$ The economy consists of N sectors indexed by $i, j \in \mathcal{N} = \{1, \ldots, N\}$.

2.1 Production and Pricing

Final and sectoral aggregation. The economy produces a final good Y_t which is a Cobb-Douglas aggregate of sectoral bundles $\{Y_{it}\}_{i \in \mathcal{N}}$:

$$Y_t = \prod_{i \in \mathcal{N}} (Y_{it}/\alpha_{it})^{\alpha_i}, \qquad \sum_{i \in \mathcal{N}} \alpha_i = 1, \tag{1}$$

with associated consumer price index $P_t^C = P_{it}^{\alpha_i}$, where P_{it} is the price index of sector i. Each sectoral bundle is a CES aggregate of differentiated varieties $f \in [0, 1]$ with elasticity of substitution $\epsilon_i > 1$:

$$Y_{it} = \left(\int_0^1 Y_{it}(f)^{\frac{\epsilon_i - 1}{\epsilon_i}} \, df \right)^{\frac{\epsilon_i}{\epsilon_i - 1}}, \tag{2}$$

yielding demand for variety f and the sectoral price index:

$$Y_{it}(f) = \left(\frac{P_{it}(f)}{P_{it}}\right)^{-\epsilon_i} Y_{it}, \qquad P_{it} = \left(\int_0^1 P_{it}(f)^{1-\epsilon_i}\, df\right)^{\frac{1}{1-\epsilon_i}}. \tag{3}$$

Technology. Each firm f in sector i operates a linear production technology:

$$Y_{it}(f) = A_{it} L_{it}(f), \tag{4}$$

where A_{it} is sector-specific productivity and $L_{it}(f)$ is labor input. Nominal marginal cost is $MC_{it} = W_{it}/A_{it}$, where W_{it} is the nominal wage in sector i.

Pricing. Firms set prices subject to Calvo frictions. In each period, a fraction $1 - \delta_i \in (0,1)$ of firms in sector i can reset their prices. The government provides a revenue subsidy of rate $\frac{1}{\epsilon_i - 1}$ to each sector, financed by lump-sum taxes, that offsets the monopolistic markup and ensures the flexible-price steady state is efficient. Under this subsidy, a re-optimizing firm chooses $P^*_{it}(f)$ to maximize:

$$\max_{P^*_{it}(f)} \mathbb{E}_t \sum_{s=0}^{\infty} \delta_i^s Q_{t,t+s} \left(\frac{P^*_{it}(f)}{P_{it+s}}\right)^{-\epsilon_i} Y_{it+s} \left[\frac{\epsilon_i}{\epsilon_i - 1} P^*_{it}(f) - MC_{it+s}\right], \tag{5}$$

where $Q_{t,t+s} = \beta^s (C_{t+s}/C_t)^{-\gamma} (P^C_t / P^C_{t+s})$ is the stochastic discount factor.

2.2 Households

A representative family consists of a unit mass of workers distributed across sectors. Let N_{it} denote the fraction of workers in sector i at the start of period t, with $\sum_{i \in \mathcal{N}} N_{it} = 1$. Each worker in sector i supplies L_{it} hours during period t. The distribution $\{N_{it}\}$ evolves through a reallocation process described in Section 2.3.

Preferences. The family maximizes:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \sum_{i \in \mathcal{N}} N_{it}\, \zeta_i \frac{L_{it}^{1+\varphi}}{1+\varphi} - \Psi_t\right], \tag{6}$$

where $\beta \in (0,1)$ is the discount factor, $\gamma > 0$ is the inverse of the elasticity of intertemporal substitution, $\varphi > 0$ is the inverse Frisch elasticity, $\zeta_i > 0$ are sector-specific disutility shifters, and Ψ_t is the cost of labor reallocation incurred at period t (defined in Section 2.3).

Budget constraint. The family faces the nominal budget constraint:

$$P_t^C C_t + B_t \leq (1 + i_{t-1}) B_{t-1} + \sum_{i \in \mathcal{N}} W_{it} L_{it} N_{it} + \Pi_t + T_t, \tag{7}$$

where B_t are one-period nominal bonds paying gross interest $1 + i_t$ set by the central bank, Π_t are aggregate profits, and T_t are net lump-sum transfers from the government.

Optimality conditions. Maximizing (6) subject to (7), taking $\{N_{it}\}$ as given, yields the intratemporal labor supply condition for each sector i:

$$\frac{W_{it}}{P_t^C} = \zeta_i L_{it}^{\varphi} C_t^{\gamma}, \tag{8}$$

and the intertemporal Euler equation:

$$1 = \beta (1 + i_t) \, \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{P_t^C}{P_{t+1}^C} \right]. \tag{9}$$

2.3 Labor Reallocation

We now describe how workers move across sectors between periods. At the end of period t, after production and consumption decisions have been made, the representative family coordinates the reallocation of its workers for the start of period $t + 1$.[1]

Reallocation cost. The per-period reallocation cost in the family's objective (6) takes the form:

$$\Psi_t = \sum_{i \in \mathcal{N}} N_{it} \, \psi_i \left(\{\mu_t^{ij}\}_{j \in \mathcal{N}} \right), \tag{10}$$

where $\psi_i(\cdot)$ is a sector-specific cost function that is strictly convex in the transition probabilities $\{\mu_t^{ij}\}$ satisfying $\mu_t^{ij} \geq 0, \sum_{j \in \mathcal{N}} \mu_t^{ij} = 1$. We adopt the following parametric specification, standard in the dynamic discrete choice literature (Anderson et al., 1988; Artuç et al., 2010; Caliendo et al., 2019):

$$\psi_i \left(\{\mu_t^{ij}\}_j \right) = \sum_{j \in \mathcal{N}} \mu_t^{ij} \tau^{ij} + \frac{1}{\theta} \sum_{j \in \mathcal{N}} \mu_t^{ij} \log \mu_t^{ij}, \tag{11}$$

where $\tau^{ij} \geq 0$ are bilateral reallocation costs (we normalize $\tau^{ii} = 0$). This aggregate penalty is equivalent to a decentralized setting in which individual workers draw i.i.d. type-I ex-

[1] The timing is: (1) period t begins with employment distribution $\{N_{it}\}$ and last period's prices $\{P_{it-1}\}$ given; (2) shocks are realized, firms set prices under Calvo frictions, and workers supply hours L_{it}; (3) at the end of period t, workers choose which sector to enter for period $t + 1$; (4) this determines $\{N_{it+1}\}$.

treme value preference shocks with dispersion $1/\theta$ (Anderson et al., 1992). The parameter $\theta > 0$ governs the elasticity of labor reallocation: as θ falls, flows become less responsive to continuation-value differentials. In the limit $\theta \to 0$, sectoral employment shares do not respond to shocks at first order. Conversely, as $\theta \to \infty$, the family sorts workers deterministically toward high-value destinations, approaching a frictionless benchmark.

Value functions and optimal flows. The family chooses the flows $\{\mu_t^{ij}\}$ to maximize welfare. Let V_{it} denote the marginal continuation value to the family of having an additional worker in sector i at time t. Under the functional form for ψ_i, optimal transition probabilities take the logit form:

$$\mu_t^{ij} = \frac{\exp\left(\theta[\beta\,\mathbb{E}_t V_{jt+1} - \tau^{ij}]\right)}{\sum_{j'\in\mathcal{N}} \exp\left(\theta[\beta\,\mathbb{E}_t V_{j't+1} - \tau^{ij'}]\right)}. \tag{12}$$

Substituting these optimal flows back into the family's objective yields a recursive expression for the marginal value of a worker:

$$V_{it} = U_{it}^{\text{flow}} + \frac{1}{\theta}\log\left(\sum_{j\in\mathcal{N}} \exp\left(\theta[\beta\mathbb{E}_t V_{jt+1} - \tau^{ij}]\right)\right), \tag{13}$$

where the flow utility from working in sector j at time t is:

$$U_{jt}^{\text{flow}} \equiv \frac{W_{jt}}{P_t^C} C_t^{-\gamma} L_{jt} - \zeta_j \frac{L_{jt}^{1+\varphi}}{1+\varphi}. \tag{14}$$

Using the intratemporal labor supply condition (8), this simplifies to:

$$U_{jt}^{\text{flow}} = \zeta_j \frac{\varphi}{1+\varphi} L_{jt}^{1+\varphi}. \tag{15}$$

Steady-state balance. We assume bilateral reallocation costs satisfy:

Assumption 1 (Quasi-symmetric reallocation costs)**.** *The bilateral reallocation costs satisfy $\tau^{ii} = 0$ and admit a decomposition $\tau^{ij} = \tau_i^{\text{out}} + \tau_j^{\text{in}} + \hat{\tau}^{ij}$ for $i \neq j$, where $\hat{\tau}^{ij} = \hat{\tau}^{ji}$ for all i, j.*

Quasi-symmetry nests symmetric costs ($\tau^{ij} = \tau^{ji}$) and additive origin-destination costs ($\tau^{ij} = \tau_1^i + \tau_2^j$); it is imposed both to reduce the dimensionality of cost parameters and to simplify the equilibrium system (e.g., Artuç et al., 2010; Allen et al., 2020; Seo and Oh, 2025). A corollary is that the steady-state transition matrix $\bar{\mu} = \{\bar{\mu}^{ij}\}$ satisfies the detailed balance condition:

$$\bar{N}_i \bar{\mu}^{ij} = \bar{N}_j \bar{\mu}^{ji}, \qquad \forall\, i, j \in \mathcal{N}, \tag{16}$$

where $\bar{N}_i$ is the steady-state employment share. Detailed balance is used in Section 4 to symmetrize the transition matrix and decompose the dynamic system into independent components.

Sufficient statistics. While equations (12) and (13) provide the microfoundations for labor reallocation, the subsequent analysis of the linearized equilibrium (Section 2.5) reveals that the specific bilateral costs τ^{ij} and micro-parameters θ, ζ do not enter the dynamics independently. Instead, the labor block is summarized by two objects: the *steady-state transition matrix* $\bar{\mu} \equiv \{\bar{\mu}^{ij}\}$ which discipline baseline worker flows and satisfies detailed balance under assumption 1; and the mobility elasticity $\phi \equiv \varphi\theta\zeta\bar{L}^{1+\varphi}$, a composite parameter that governs how continuation-value differentials tilt aggregate flows.

2.4 Equilibrium

Market clearing. In equilibrium, the goods market for each variety and the aggregate economy must clear:

$$Y_{it}(f) = C_{it}(f), \quad Y_t = C_t, \tag{17}$$

where C_t is household consumption. Combined with Cobb-Douglas demand, this implies that nominal sectoral output equals the expenditure share of total nominal consumption:

$$P_{it}Y_{it} = \alpha_i P_t^C C_t. \tag{18}$$

Labor market clearing in each sector requires:

$$\int_0^1 L_{it}(f)\, df = N_{it}L_{it}. \tag{19}$$

The government finances the output subsidies through lump-sum taxes, balancing its budget each period:

$$T_t + \sum_{i\in\mathcal{N}} \frac{1}{\epsilon_i - 1} P_{it}Y_{it} = 0. \tag{20}$$

The central bank sets the nominal interest rate i_t according to a policy rule specified below.

Definition 1 (Equilibrium)**.** *Given an initial employment distribution $\{N_{i0}\}$ and a sequence of productivity shocks $\{A_{it}\}$, an equilibrium is a collection of prices $\{P_{it}(f), P_{it}, P_t^C, W_{it}, i_t\}$, quantities $\{C_t, Y_{it}, L_{it}(f), L_{it}, B_t\}$, employment shares $\{N_{it}\}$, transition probabilities $\{\mu_t^{ij}\}$, and value functions $\{V_{it}\}$ such that: (i) households optimize consumption, savings, labor supply, and reallocation; (ii) firms optimize pricing subject to Calvo constraints; (iii) all markets clear and the government budget constraint*

(20) *holds; and (iv) the central bank follows its policy rule.*

Steady state. We impose that steady-state flow surplus is equalized across sectors: $\zeta_i \bar{L}_i^{1+\varphi}$ is constant across i. This is the natural interior condition for the logit reallocation problem: otherwise, as $\theta \to \infty$, workers would concentrate in the highest-surplus sector. We achieve this by normalizing $\bar{L}_i = \bar{L}$ and $\zeta_i = \zeta$ for all i.[2] Combining steady-state labor supply (8) with goods market clearing (18) gives $\alpha_i \bar{C}^{1-\gamma} = \zeta \bar{N}_i \bar{L}^{1+\varphi}$, so equal surplus implies $\bar{N}_i = \alpha_i$: steady-state employment shares equal consumption expenditure shares.

2.5 Log-Linearized Equilibrium

Log-linearization of the equilibrium conditions yields the following system; derivations are in Appendix A.

Notation. For any variable X_t, we write $x_t \equiv \log X_t - \log \bar{X}$ for its log-deviation from steady state. We use the following conventions throughout:

- *Natural (flexible-price) levels*: x_t^{nat} denotes the value under flexible prices.
- *Gaps*: $\tilde{x}_t \equiv x_t - x_t^{\text{nat}}$ denotes the deviation from the natural level.
- *Relative prices*: $\hat{p}_{it} \equiv p_{it} - p_t^C$ is the log sectoral price relative to the CPI.
- *Relative price distortion*: $\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}$, where $\hat{a}_{it} \equiv a_{it} - \sum_j \alpha_j a_{jt}$. In the static economy this equals $\hat{p}_{it} - \hat{p}_{it}^{\text{nat}}$ (Lemma 2). In the dynamic economy with predetermined n_{it}, the true flexible-price relative price also depends on inherited employment (Lemma 4), so χ_{it} measures the deviation from the *static* natural benchmark throughout; the distinction is discussed after Lemma 4.

We write $\overset{(1)}{=}$ for equalities that hold up to a first-order approximation, and $\overset{(2)}{=}$ for second-order. Plain $=$ denotes exact relationships.

The sectoral New Keynesian Phillips curve is:

$$\pi_{it} \overset{(1)}{=} \kappa_i \left[mc_{it} - p_{it} \right] + \beta \, \mathbb{E}_t \pi_{it+1}, \tag{21}$$

where $\pi_{it} = p_{it} - p_{it-1}$ is sectoral inflation and $\kappa_i \equiv (1-\delta_i)(1-\beta\delta_i)/\delta_i$. Using the production technology (4), labor supply (8), goods market clearing ($y_{it} = a_{it} + n_{it} + l_{it}$), and sectoral demand

[2]The normalization $\bar{L}_i = \bar{L}$ is without loss of generality: N_{it} measures the mass of workers in sector i, and L_{it} can be interpreted as either per-worker hours or the fraction of workers who supply labor (Galí, 2011). The condition $\zeta_i = \zeta$ then ensures equal surplus.

$(y_{it} - y_t = -\hat{p}_{it})$, the marginal cost is:

$$mc_{it} - p_{it} \overset{(1)}{=} (\gamma + \varphi) y_t - (1 + \varphi)(\hat{p}_{it} + a_{it}) - \varphi n_{it}. \tag{22}$$

The IS curve is:

$$\tilde{y}_t \overset{(1)}{=} \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t \pi^C_{t+1} - r^{\text{nat}}_t\right), \tag{23}$$

where $\pi^C_t = \sum_i \alpha_i \pi_{it}$ is CPI inflation and $r^{\text{nat}}_t = -\ln \beta + \gamma(\mathbb{E}_t y^{\text{nat}}_{t+1} - y^{\text{nat}}_t)$ is the natural real rate, characterized in Sections 3–4. Relative prices evolve as:

$$\hat{p}_{it} = \hat{p}_{it-1} + \pi_{it} - \pi^C_t. \tag{24}$$

The labor reallocation block links employment dynamics to forward-looking value functions. Using bold for N-vectors ($\mathbf{n}_t$, $\mathbf{v}_t$, $\mathbf{u}_t$) and normalizing value-function and flow-utility deviations by $\varphi\zeta\bar{L}^{1+\varphi}$, the reallocation block is:

$$\mathbf{n}_{t+1} \overset{(1)}{=} \bar{\mu}\, \mathbf{n}_t + \phi\beta(I - \bar{\mu}^2)\, \mathbb{E}_t[\mathbf{v}_{t+1}], \tag{25}$$

$$\mathbf{v}_t \overset{(1)}{=} \mathbf{u}_t + \beta\bar{\mu}\, \mathbb{E}_t[\mathbf{v}_{t+1}], \tag{26}$$

$$u_{it} \overset{(1)}{=} y_t - \hat{p}_{it} - a_{it} - n_{it}, \tag{27}$$

where $\phi \equiv \varphi\theta\zeta\bar{L}^{1+\varphi}$ is the mobility elasticity. Here $\mathbf{v}_t$ and $\mathbf{u}_t$ are normalized deviations from steady-state (not log-deviations), with $v_{it} =\equiv (V_{it} - \bar{V}_i)/(\varphi\zeta\bar{L}^{1+\varphi})$ and u_{it} defined analogously. The aggregate constraint $\sum_i \alpha_i n_{it} = 0$ holds throughout.[3]

Finally, the central bank sets the nominal interest rate according to:

$$i_t = \psi_\pi \pi_t(\omega) + \psi_y \tilde{y}_t, \tag{28}$$

for some inflation index $\pi_t(\omega) = \sum_i \omega_i \pi_{it}$ with weights summing to one.

Discussion. The system (21)–(28) nests two benchmarks: as $\phi \to 0$, employment shares are fixed and the model reduces to an economy with sector-specific labor; as $\phi \to \infty$, flows equalize utilities across sectors, recovering a single-labor-market economy (e.g., Rubbo, 2023).

We analyze this system in two steps. Section 3 studies a static formulation in which n_{it} adjusts within the period, collapsing the reallocation block into a single sufficient statistic. Section 4 studies the dynamic formulation in which n_{it} is predetermined, and where ϕ and $\bar{\mu}$ shapes the persistence of sectoral inflation and the structure of optimal monetary policy.

[3]Combining steady-state labor supply (8) with goods market clearing (18) gives $\zeta\bar{L}^{1+\varphi} = \bar{C}^{1-\gamma}$, so $\phi = \varphi\theta\bar{C}^{1-\gamma}$.

3 Static Economy: Mobility Elasticity as Sufficient Statistic

We first analyze the case where labor reallocates within the period. This collapses the system into a single sufficient statistic. This provides analytical intuition for how mobility shapes the sectoral Phillips curve and the central bank's optimal inflation-targeting weights.

3.1 The static limit

Suppose workers choose sectors each period and today's choice does not constrain tomorrow's feasible set. The continuation value and bilateral costs are absorbed into the steady-state shares $\bar{N}_i$, so workers choose based on current flow utility alone. The transition equation (12) reduces to:

$$N_{it} = \frac{\bar{N}_i \exp(\theta U_{it}^{\text{flow}})}{\sum_{j \in \mathcal{N}} \bar{N}_j \exp(\theta U_{jt}^{\text{flow}})}. \tag{29}$$

where $U_{it}^{\text{flow}} = \zeta \frac{\varphi}{1+\varphi} L_{it}^{1+\varphi}$ from (15). Employment $\{N_{it}\}$ is now a jump variable that adjusts contemporaneously to shocks. Log-linearizing (29) around the symmetric steady state:

$$n_{it} \overset{(1)}{=} \phi(l_{it} - l_t^{\alpha}), \qquad l_t^{\alpha} \equiv \sum_i \alpha_i l_{it}, \tag{30}$$

Workers flow toward sectors with above-average hours (and hence above-average wages) with elasticity ϕ. The constraint $\sum_i \alpha_i n_{it} = 0$ is satisfied by construction.

Substituting (30) into the Phillips curve aggregates the extensive margin (reallocation) and intensive margin (hours) into a single labor supply response:

Lemma 1 (Extensive Margin Aggregation)**.** *Under* (30)*, aggregate output satisfies* $y_t = \sum_i \alpha_i a_{it} + \sum_i \alpha_i (n_{it} + l_{it})$*. Since* $\sum_i \alpha_i n_{it} = 0$*:*

$$\tilde{y}_t = \sum_i \alpha_i \tilde{l}_{it} = \tilde{l}_t^{\alpha}.$$

The output gap depends only on the weighted average of intensive-margin labor gaps, not on employment reallocation.

3.2 Flexible-Price Allocation

We characterize the flexible-price allocation. Since the output subsidy offsets the monopolistic markup, the flexible-price competitive equilibrium is efficient. In the static limit, the dynamic reallocation friction collapses to a static relative-entropy penalty that anchors employment to the steady-state baseline shares $\{\bar{N}_i\}$, and the planner's problem reduces to a per-period choice

of $\{L_{it}, N_{it}\}$ to maximize

$$\frac{C^{1-\gamma}}{1-\gamma} - \sum_i \zeta N_{it} \frac{L_{it}^{1+\varphi}}{1+\varphi} - \frac{1}{\theta} \sum_i N_{it} \log \frac{N_{it}}{\bar{N}_i} \tag{31}$$

subject to $C_t = \prod_{i \in \mathcal{N}} (A_{it} N_{it} L_{it})^{\alpha_i}$ and the employment constraint $\sum_{i \in \mathcal{N}} N_{it} = 1$. The final term in (31) is exactly the penalty that rationalizes the decentralized logit choice (29).

The first-order condition for L_{it} gives:

$$\alpha_i C^{1-\gamma} = \zeta N_{it} L_{it}^{1+\varphi}. \tag{32}$$

The first-order condition for N_{it} reproduces the decentralized extensive margin condition; taking logs, it yields (30).

Taking logs of (32) gives $(1-\gamma) y_t = n_{it} + (1+\varphi) l_{it}$. Weighting by α_i and summing:

$$l_t^{\alpha} = \frac{1-\gamma}{1+\varphi} y_t. \tag{33}$$

Substituting back, the intensive margin condition becomes $(1+\varphi)(l_{it} - l_t^{\alpha}) = -n_{it}$. Combined with $n_{it} = \phi(l_{it} - l_t^{\alpha})$, this implies $l_{it} = l_t^{\alpha}$ and $n_{it} = 0$ for all i: the planner equalizes real wage (and hours) across sectors and does not reallocate workers away from steady state.

Lemma 2 (Natural Allocation Invariance)**.** *The flexible-price planner allocation features uniform sectoral hours and zero reallocation:*

$$l_{it}^{nat} = \frac{1-\gamma}{1+\varphi} y_t^{nat}, \qquad n_{it}^{nat} = 0.$$

Natural output and natural relative prices are:

(i) $y_t^{nat} = \frac{1+\varphi}{\gamma+\varphi} \sum_i \alpha_i a_{it}$,

(ii) $\hat{p}_{it}^{nat} = \sum_j \alpha_j a_{jt} - a_{it}$,

and are both independent of the mobility parameter ϕ.

Proof. Natural output follows from substituting $l_t^{\alpha} = \frac{1-\gamma}{1+\varphi} y_t$ and $\sum_i \alpha_i n_{it} = 0$ into $y_t = \sum_i \alpha_i a_{it} + \sum_i \alpha_i (n_{it} + l_{it})$ and solving. For natural relative prices, goods market clearing gives $\hat{p}_{it}^{\text{nat}} = y_t^{\text{nat}} - a_{it} - l_{it}^{\text{nat}} - n_{it}^{\text{nat}}$. Substituting $n_{it}^{\text{nat}} = 0$ and $l_{it}^{\text{nat}} = \frac{1-\gamma}{1+\varphi} y_t^{\text{nat}}$ yields $\hat{p}_{it}^{\text{nat}} = \frac{\gamma+\varphi}{1+\varphi} y_t^{\text{nat}} - a_{it} = \sum_j \alpha_j a_{jt} - a_{it}$. Neither expression involves ϕ. □

Under flexible prices, the planner achieves the efficient allocation solely through the intensive margin (l_{it}^{nat}), leaving no incentive for reallocation. Therefore the mobility parameter ϕ is irrelevant for the natural allocation. Mobility matters for welfare and dynamics only once sticky prices drive the economy away from this benchmark.

3.3 Phillips Curve with Mobility Frictions

We now show how labor mobility enters the Phillips curve through a single sufficient statistic.

Marginal cost in gaps. From log-linearized marginal cost (22) and Lemma 2, the real marginal cost in gaps is:

$$mc_{it} - p_{it} = \gamma \tilde{y}_t + \varphi \tilde{l}_{it} - \chi_{it}, \tag{34}$$

where $\chi_{it} \equiv \hat{p}_{it} - \hat{p}_{it}^{\text{nat}}$ is the relative price distortion and we have used $mc_{it}^{\text{nat}} = p_{it}^{\text{nat}}$ (from efficient subsidy). ′

Margin decomposition. Total sectoral labor gaps satisfy $\tilde{h}_{it} = \tilde{n}_{it} + \tilde{l}_{it}$, and goods market clearing requires $\tilde{h}_{it} = \tilde{y}_t - \chi_{it}$. Combined with (30), a unit relative price distortion χ_{it} induces a total labor adjustment of -1, split as:

$$\text{intensive:} \quad \tilde{l}_{it} = \tilde{y}_t - \frac{1}{1+\phi}\chi_{it}, \tag{35}$$

$$\text{extensive:} \quad \tilde{n}_{it} = -\frac{\phi}{1+\phi}\chi_{it}. \tag{36}$$

Higher ϕ shifts the adjustment from hours to worker flows. Substituting (35) into the marginal cost gap (34):

$$mc_{it} - p_{it} = (\gamma + \varphi)\,\tilde{y}_t - \sigma\,\chi_{it}, \qquad \sigma \equiv 1 + \frac{\varphi}{1+\phi} \in [1,\, 1+\varphi). \tag{37}$$

The labor supply margin contributes φ to the marginal cost response, but only a fraction $1/(1+\phi)$ of sectoral adjustment occurs through hours; worker flows absorb the rest. The effective response is therefore $1 + \varphi/(1+\phi) = \sigma$, not $1+\varphi$. Full mobility ($\phi \to \infty$) gives $\sigma = 1$; segmented markets ($\phi = 0$) gives $\sigma = 1 + \varphi$.

Sectoral NKPC. Labor reallocation is static, but price setting remains forward-looking under Calvo. We characterize the NKPC by treating the expectations $\{\mathbb{E}_t \pi_{it+1}\}$ as given; the full dynamic solution is in Section 4. The relative price distortion evolves as:

$$\chi_{it} = (\pi_{it} - \pi_t^C) + \chi_{i,t-1} + \Delta\hat{a}_{it}, \tag{38}$$

where $\hat{a}_{it} \equiv a_{it} - \sum_j \alpha_j a_{jt}$. Substituting (37) into (21) and solving for CPI inflation (Appendix A):

$$\begin{aligned}\pi_{it} = b_i \tilde{y}_t - \frac{\bar{\kappa}_i}{1-\bar{\kappa}^\alpha}\left[(1-\bar{\kappa}^\alpha)(\chi_{i,t-1} + \Delta\hat{a}_{it}) + \textstyle\sum_j \alpha_j \bar{\kappa}_j (\chi_{j,t-1} + \Delta\hat{a}_{jt})\right] \\ + \frac{\beta}{1+\sigma\kappa_i}\mathbb{E}_t \pi_{it+1} + \frac{\beta\bar{\kappa}_i}{1-\bar{\kappa}^\alpha}\textstyle\sum_j \frac{\alpha_j}{1+\sigma\kappa_j}\mathbb{E}_t \pi_{jt+1}\end{aligned} \tag{39}$$

where the adjusted Calvo coefficient and sectoral output gap slopes are:

$$\bar{\kappa}_i \equiv \frac{\sigma\kappa_i}{1+\sigma\kappa_i} \in (0,1), \qquad \bar{\kappa}^\alpha \equiv \sum_i \alpha_i \bar{\kappa}_i, \tag{40}$$

$$b_i \equiv \frac{\gamma+\varphi}{\sigma}\frac{\bar{\kappa}_i}{1-\bar{\kappa}^\alpha} = \frac{(\gamma+\varphi)\,\kappa_i}{(1+\sigma\kappa_i)(1-\bar{\kappa}^\alpha)}. \tag{41}$$

The coefficient $\bar{\kappa}_i$ measures how responsive sector-i inflation is to contemporaneous cost pressures, accoiunting for the feedback between mobility and relative prices. It is increasing in both κ_i and σ.

Corollary 1. *When $\kappa_i = \kappa$ for all i, $1-\bar{\kappa}^\alpha = (1+\sigma\kappa)^{-1}$, so $b_i = (\gamma+\varphi)\kappa$ for all i, independent of σ. Mobility then affects the Phillips curve only through cost-push shocks and relative price dynamics, not through the output gap slope.*

3.4 Monetary Policy Problem and Optimal Rule

Because the efficient steady state is supported by sectoral subsidies, the first-order terms in the welfare expansion vanish, justifying a linear-quadratic approach.

Proposition 1 (Discretionary LQ Problem)**.** *In each period t, the central bank chooses the output gap $\tilde{y}_t$:*

$$\min_{\tilde{y}_t} \mathcal{L}_t \overset{(2)}{=} \frac{\gamma+\varphi}{2}\tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i}\pi_{it}^2 + \frac{\sigma}{2}\sum_i \alpha_i \chi_{it}^2, \tag{42}$$

subject to the sectoral Phillips curve

$$\pi_{it} = b_i \tilde{y}_t + u_{it}, \tag{43}$$

where $\sigma = 1 + \frac{\varphi}{1+\phi}$ and χ_{it} is the relative price distortion (38).

Proof. Derived in Appendix B. □

The three terms in (42) identify distinct economic inefficiencies. The first two terms—the aggregate output gap and within-sector price dispersion—are standard. The third term penalizes cross-sector relative price distortions, which misallocate expenditure and labor. Under

segmented markets ($\sigma = 1 + \varphi$), relative price distortions are fully absorbed by hours, generating large welfare costs. As mobility increases ($\sigma \to 1$), worker flows absorb these distortions, reducing the welfare penalty to residual expenditure misallocation (Rubbo, 2023). That σ appears both as the marginal cost slope on relative price distortions (37) and as the welfare weight here is reflect the same margin decomposition (35)–(36), making σ a *sufficient statistic* for sectoral reallocation in the static problem.

The central bank's only instrument is $\tilde{y}_t$. From (43), aggregate demand shifts sector-i inflation by b_i, and by (38) relative price distortions by $(b_i - b^C)$, where $b^C \equiv \sum_j \alpha_j b_j$. All other terms (initial price wedges, productivity shocks, and inflation expectations) are embedded in u_{it} and taken as given.

Before solving the full problem, we establish a benchmark. Is there an inflation index whose stabilization closes the output gap, regardless of labor mobility?

Lemma 3 (Output-Gap-Targeting Inflation Index)**.** *Define the α/κ-weighted inflation index $\pi_t^R \equiv \sum_i \frac{\alpha_i}{\kappa_i}\pi_{it}$. Then*

$$\sum_i \frac{\alpha_i}{\kappa_i}(\pi_{it} - \beta \mathbb{E}_t \pi_{it+1}) = (\gamma + \varphi)\tilde{y}_t. \tag{44}$$

If policy implements $\pi_t^R = 0$ for all t, then $\tilde{y}_t = 0$.

Proof. Summing the NKPC (21) with weights α_i/κ_i gives $\sum_i \frac{\alpha_i}{\kappa_i}(\pi_{it} - \beta\, \mathbb{E}_t \pi_{it+1}) = \sum_i \alpha_i (mc_{it} - p_{it})$. From (34), $mc_{it} - p_{it} = \gamma \tilde{y}_t + \varphi \tilde{l}_{it} - \chi_{it}$. Using $\chi_{it} = \tilde{y}_t - \tilde{h}_{it}$ and $\tilde{h}_{it} = \tilde{n}_{it} + \tilde{l}_{it}$, then substituting the static reallocation condition $\tilde{n}_{it} = \phi(\tilde{l}_{it} - \tilde{y}_t)$:

$$mc_{it} - p_{it} = (\gamma - 1 - \phi)\, \tilde{y}_t + (1 + \varphi + \phi)\, \tilde{l}_{it}.$$

Weighting by α_i and summing:

$$\sum_i \alpha_i (mc_{it} - p_{it}) = (\gamma - 1 - \phi)\, \tilde{y}_t + (1 + \varphi + \phi)\, \tilde{l}_t^{\alpha} = (\gamma + \varphi)\, \tilde{y}_t,$$

where the last equality uses $\tilde{l}_t^{\alpha} = \tilde{y}_t$ (Lemma 1). All terms involving ϕ cancel. □

This result extends the aggregation result of Rubbo (2023) to economies with frictional labor mobility: because the intensive and extensive margin adjustments offset upon aggregation, all mobility terms cancel, and if the central bank stabilizes π_t^R, the output gap would be closed. However, $\pi_t^R = 0$ does not imply $\pi_{it} = 0$ or $\chi_{it} = 0$ for each i, so it leaves the dispersion terms in (42) nonzero. Optimal policy must therefore trade off output gap stabilization against cross-sector misallocation:

Proposition 2 (Optimal Discretionary Targeting Rule)**.** *The optimal discretionary policy satisfies*

$$0 = (\gamma + \varphi)\,\tilde{y}_t + \sum_i \alpha_i \omega_i\, \pi_{it} + c_t, \tag{45}$$

with sectoral inflation weights

$$\omega_i = \underbrace{\frac{\epsilon_i}{\kappa_i}\, b_i}_{\textit{within-sector (Calvo)}} + \underbrace{\sigma(b_i - b^C)}_{\textit{cross-sector (dispersion)}}, \tag{46}$$

and cost-push term $c_t = \sigma \sum_i \alpha_i (b_i - b^C)(\chi_{i,t-1} + \Delta\hat{a}_{it})$.

Proof. The central bank's instrument is $\tilde{y}_t$. From (43), $\partial\pi_{it}/\partial\tilde{y}_t = b_i$. From (38), $\partial\chi_{it}/\partial\tilde{y}_t = b_i - b^C$, where $b^C \equiv \sum_j \alpha_j b_j$. Differentiating (42) with respect to $\tilde{y}_t$:

$$0 = (\gamma + \varphi)\,\tilde{y}_t + \sum_i \frac{\alpha_i \epsilon_i}{\kappa_i}\, \pi_{it}\, b_i + \sigma \sum_i \alpha_i\, \chi_{it}\, (b_i - b^C).$$

Substituting $\chi_{it} = \chi_{i,t-1} + \pi_{it} - \pi_t^C + \Delta\hat{a}_{it}$ and collecting the period-t inflation terms (using π_{is} for $s < t$ as given under discretion) yields (45) with $\omega_i = \frac{\epsilon_i}{\kappa_i} b_i + \sigma(b_i - b^C)$ and $c_t = \sigma \sum_i \alpha_i (b_i - b^C)(\chi_{i,t-1} + \Delta\hat{a}_{it})$. □

The weight ω_i has two components. The Calvo component, $\frac{\epsilon_i}{\kappa_i} b_i$, is standard and large for sticky sectors: high ϵ_i/κ_i means that inflation in sector i generates large within-sector price dispersion. The dispersion channel, $\sigma(b_i - b^C)$, gives positive (negative) weight to sectors whose Phillips curve is steeper (flatter) than average. These are the sectors where a unit change in $\tilde{y}_t$ produces the largest relative price movement. This component is scaled by σ: under segmented markets ($\sigma = 1 + \varphi$), the dispersion channel is large; under mobile labor ($\sigma \to 1$), workers absorb relative price distortions and the dispersion channel shrinks.

When $\kappa_i = \kappa$ for all i, $b_i = b^C$ and the dispersion channel vanishes. The optimal index then weights sectoral inflation solely through within-sector dispersion costs, proportional to ϵ_i. If additionally $\epsilon_i = \epsilon$, optimal policy reduces to stabilizing the CPI, consistent with Corollary 1 under constant κ_i. More generally, the optimal policy puts weight on both sticky and flexible sectors—sticky sectors through the Calvo channel, and flexible sectors (sectors with high b_i) through the dispersion channel.

Illustration. To gauge the quantitative importance of the two components, consider a two-sector calibration. We approximate the distribution of sectoral price rigidities documented in Pastén et al. (2024) by grouping sectors into a flexible sector (food and energy, $\alpha_1 = 0.5$, average monthly adjustment frequency of 40%) and a sticky sector (remaining goods, $\alpha_2 = 0.5$, average monthly frequency of 12%), implying quarterly Calvo parameters $\delta_1 = 0.2$ ($\kappa_1 \approx 3.2$)

	Segmented Markets ($\sigma = 3$)			Integrated Markets ($\sigma = 1$)		
	Calvo	Disp.	Total (ω_i)	Calvo	Disp.	Total (ω_i)
Flexible ($i = 1$)	7.0	+2.1	**9.1**	11.1	+1.6	**12.7**
Sticky ($i = 2$)	41.0	−2.1	**38.9**	36.9	−1.6	**35.3**

Table 1: Two-sector experiment with segmented vs integrated markets.

and $\delta_2 = 0.6$ ($\kappa_2 \approx 0.27$). We set $\gamma = 1$, $\varphi = 2$, $\beta = 0.99$, and $\epsilon_i = 8$ for both sectors, following Rubbo (2023).

To illustrate, we compare the optimal targeting weights under segmented markets ($\sigma = 1 + \varphi = 3$) and fully integrated markets ($\sigma \to 1$). Under segmented markets, the output gap slopes are $b_1 \approx 2.8$ (flexible) and $b_2 \approx 1.4$ (sticky). Under integrated markets, the flexible sector can freely absorb workers, steepening its slope to $b_1 \approx 4.5$, while the sticky sector flattens slightly to $b_2 \approx 1.2$.

Table 1 presents the optimal weights. In both regimes, the expenditure-weighted inflation penalty $\sum \alpha_i \omega_i$ exceeds the output gap weight $\gamma + \varphi = 3$ byt an order of magnitude, so the planner prioritizes inflation stabilization. The Calvo channel dominates in both regimes, prioritizing the sticky sector. ut the weights shift with mobility. Under integrated markets, the flexible sector's steeper slope b_1 significantly raises the Calvo penalty for flexible-sector inflation ($\frac{\epsilon_1}{\kappa_1} b_1$). Consequently, the optimal index shifts *toward* the flexible sector, with the sticky-to-flexible weight ratio falling from roughly 4.3x under segmented markets to 2.8x under integrated markets. A central bank that assumes integrated markets when labor is in fact segmented will under-weight sticky-sector inflation. When labor cannot reallocate easily, the central bank must tilt its focus heavily toward stabilizing sticky prices to prevent severe misallocation.

Turning to dynamics. When employment is predetermined, labor misallocation cannot be corrected within the period, raising the intertemporal cost of relative price distortions. But predetermined employment also creates a new policy margin absent from the static model: by tilting relative prices today, the planner can redirect worker flows and improve tomorrow's employment distribution. Section 4 formalizes this tradeoff.

4 Dynamic Economy: Flows as Sufficient Statistic

We now turn to the general case in which labor reallocation is gradual: employment shares $\{N_{it}\}$ are predetermined state variables that evolve over time through worker transitions. This generates richer dynamics, but the entire micro reallocation structure enters through just two objects, the mobility elasticity ϕ and the steady-state transition matrix $\bar{\mu}$. We first analyze the

case of uniform price stickiness, where a spectral decomposition yields closed-form persistence results, then introduce heterogeneous stickiness and characterize optimal policy.

4.1 Timing and the Reallocation Block

In the dynamic model, employment shares $\mathbf{n}_t$ are predetermined: workers choose transitions at the end of period t and arrive in their new sector at $t+1$. Unlike the static economy, $\mathbf{n}_t$ can no longer be substituted out of marginal cost using a contemporaneous reallocation condition. Instead, the path of employment is governed by the forward-looking Bellman recursion (26). Iterating forward:

$$\mathbf{v}_t = \sum_{s=0}^{\infty} \beta^s \bar{\mu}^s \, \mathbb{E}_t[\mathbf{u}_{t+s}]. \tag{47}$$

The continuation value in each sector reflects the entire expected path of flow utilities, discounted by both β and $\bar{\mu}$. A sector that is temporarily depressed but expected to recover retains workers through high v_{it}, even if current flow utility u_{it} is low.

In the sectoral NKPC (22), employment enters with coefficient $-\varphi\kappa_i$, replacing the sufficient statistic σ of the static Phillips curve 37. The effect of mobility frictions on inflation now operates through the *path* of $\mathbf{n}_t$, not a contemporaneous elasticity. Despite the rich dynamics, the linearized system has a parsimonious representation:

Proposition 3 (Dynamic Sufficient Statistics)**.** *With predetermined employment shares and logit reallocation, the linearized dynamic system depends on the labor reallocation structure only through:*

(i) the mobility elasticity $\phi = \varphi\theta\zeta\bar{L}^{1+\varphi}$, and

(ii) the steady-state transition matrix $\bar{\mu}$.

The limiting cases are: $\phi \to 0$ (segmented markets), in which $\mathbf{n}_t = \mathbf{0}$ for all t; and $\phi \to \infty$ (free mobility), in which expected continuation values equalize across sectors ($\mathbb{E}_t[\mathbf{v}_{t+1}] \propto \mathbf{1}$) and, for all $t \geq 1$, employment adjusts to offset relative price distortions ($\mathbf{n}_t = -\chi_t$ in relative terms).

Proof. Inspection of (25)–(27): the transition rates $\{\mu^{ij}\}$ enter only through $\bar{\mu}$ and ϕ. When $\phi = 0$, (25) gives $\mathbf{n}_{t+1} = \bar{\mu}\,\mathbf{n}_t$ with $\mathbf{n}_0 = \mathbf{0}$, so $\mathbf{n}_t = \mathbf{0}$.

When $\phi \to \infty$, bounded $\mathbf{n}_{t+1}$ in (25) requires $(I - \bar{\mu}^2)\mathbb{E}_t[\mathbf{v}_{t+1}] \to \mathbf{0}$. Because $\bar{\mu}$ is a stochastic matrix, the null space of $(I - \bar{\mu}^2)$ is spanned by the constant vector $\mathbf{1}$. Therefore, expected continuation values equalize across sectors: $\mathbb{E}_t[\mathbf{v}_{t+1}] = c_t\mathbf{1}$ for some scalar c_t.

Assuming perfect foresight paths following the initial shock, $\mathbf{v}_t = \mathbb{E}_{t-1}[\mathbf{v}_t] = c_{t-1}\mathbf{1}$ for $t \geq 1$. Substituting this into (26) yields $\mathbf{u}_t = (c_{t-1} - \beta c_t)\mathbf{1}$, meaning flow utilities also equalize. Because we work in relative log-deviations (where aggregate constraints imply cross-sector means are

zero), cross-sector equalization requires $\mathbf{u}_t = \mathbf{0}$ for $t \geq 1$. By (27), this implies $\mathbf{n}_t = -\chi_t$ for $t \geq 1$. At impact ($t = 0$), $\mathbf{n}_0$ is predetermined and cannot jump, so $\mathbf{u}_0$ remains un-equalized. □

In the static limit, all reallocation information collapses to the scalar $\sigma = 1 + \frac{\varphi}{1+\phi}$. In the dynamic economy, the steady-state transition matrix $\bar{\mu}$ plays an additional role: its eigenvalues govern the speed at which workers reallocate, creating an interaction between price stickiness and labor mobility. Before analyzing this interaction, we first characterize the flexible-price benchmark, which now differs from the static case because inherited employment misallocation affects sectoral hours and natural relative prices even when all prices are free to adjust.

4.2 Dynamic Flexible-Price Allocation

When employment shares are predetermined, the natural allocation differs from the static case Lemma 2): the planner can adjust the intensive margin within the period but must wait for the reallocation block to correct the extensive margin.

Lemma 4 (Dynamic Natural Allocation)**.** *The flexible-price allocation satisfies:*

(i) Natural output $y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \sum_i \alpha_i a_{it}$ *and the natural real rate* $r_t^{\text{nat}} = -\ln \beta + \gamma\, \mathbb{E}_t(y_{t+1}^{\text{nat}} - y_t^{\text{nat}})$ *are unchanged from the static case and independent of* $\mathbf{n}_t$*.*

(ii) Sectoral hours and natural relative prices depend on inherited employment:

$$l_{it}^{\text{nat}} = \bar{l}_t^{\text{nat}} - \frac{1}{1+\varphi}\, n_{it}, \tag{48}$$

$$\hat{p}_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat,static}} - \frac{\varphi}{1+\varphi}\, n_{it}. \tag{49}$$

An overstaffed sector ($n_{it} > 0$) has lower hours and a lower natural relative price.

(iii) Real marginal cost in gaps decomposes as:

$$mc_{it} - p_{it} = (\gamma + \varphi)\, \tilde{y}_t - (1+\varphi)\, \chi_{it} - \varphi\, n_{it}, \tag{50}$$

where $\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}$ *is the deviation of relative prices from their* static *natural benchmark* $\hat{p}_{it}^{nat,static} = \sum_j \alpha_j a_{jt} - a_{it}$.[4]

Proof. See Appendix C. □

[4]Note that the true dynamic natural relative price is $\hat{p}_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat,static}} - \frac{\varphi}{1+\varphi}\, n_{it}$ (part (ii)), so the deviation from the contemporaneous flexible-price benchmark is $\hat{p}_{it} - \hat{p}_{it}^{\text{nat}} = \chi_{it} + \frac{\varphi}{1+\varphi}\, n_{it}$. We retain the static-benchmark χ_{it} because it and n_{it} then enter as *independent* cost shifters in (50), making the two margins of adjustment—price gaps versus employment gaps—transparent.

Part (iii) is the key equation for what follows. In the static economy, the reallocation condition $n_{it} = -\frac{\phi}{1+\phi}\chi_{it}$ combines the last two terms into $-\sigma\chi_{it}$. With predetermined employment, χ_{it} and n_{it} enter marginal cost as independent cost shifters. Sticky prices distort the reallocation process: χ_{it} contaminates the flow-utility signals that drive worker transitions, and the resulting employment misallocation feeds back into marginal cost.

4.3 Constant κ: Reallocation Dynamics and Inflation Propagation

In this subsection, we assume homogeneous stickiness: $\kappa_i = \kappa$ for all i. Under this restriction, aggregate demand shifts marginal costs uniformly across sectors, so the system decouples into an aggregate block that monetary policy controls and a relative block it cannot influence.

Lemma 5 (Block Separation)**.** *If $\kappa_i = \kappa$ for all i, the linearized system decomposes into:*

(i) An aggregate block *in (π_t^C, y_t), equivalent to a standard one-sector New Keynesian model:*

$$\pi_t^C = \kappa[(\gamma + \varphi)\, y_t - (1 + \varphi)\, \bar{a}_t] + \beta\, \mathbb{E}_t \pi_{t+1}^C, \tag{51}$$

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \tfrac{1}{\gamma}(i_t - \mathbb{E}_t \pi_{t+1}^C - r_t^{nat}), \tag{52}$$

where $\bar{a}_t = \sum_j \alpha_j a_{jt}$, closed by the Taylor rule $i_t = \psi_\pi \pi_t^C + \psi_y \tilde{y}_t$.

(ii) A relative block *in $(\hat{\boldsymbol{\pi}}_t, \hat{\mathbf{p}}_t, \mathbf{n}_t, \hat{\mathbf{v}}_t)$ that is orthogonal to y_t at first order:*

$$\hat{\boldsymbol{\pi}}_t = -\kappa[(1 + \varphi)\, \boldsymbol{\chi}_t + \varphi\, \mathbf{n}_t] + \beta\, \mathbb{E}_t \hat{\boldsymbol{\pi}}_{t+1}, \tag{53}$$

$$\boldsymbol{\chi}_t = \boldsymbol{\chi}_{t-1} + \hat{\boldsymbol{\pi}}_t + \Delta \hat{\mathbf{a}}_t, \tag{54}$$

$$\mathbf{n}_{t+1} = \bar{\mu}\, \mathbf{n}_t + \phi\beta(I - \bar{\mu}^2)\, \mathbb{E}_t[\hat{\mathbf{v}}_{t+1}], \tag{55}$$

$$\hat{\mathbf{v}}_t = -\boldsymbol{\chi}_t - \mathbf{n}_t + \beta\bar{\mu}\, \mathbb{E}_t[\hat{\mathbf{v}}_{t+1}], \tag{56}$$

where $\boldsymbol{\chi}_t \equiv \hat{\mathbf{p}}_t + \hat{\mathbf{a}}_t$ and hats denote demeaning ($\hat{x}_{it} = x_{it} - \sum_j \alpha_j x_{jt}$).

Monetary policy (y_t) affects the aggregate block but cannot influence relative inflation or reallocation dynamics.

Proof. With $\kappa_i = \kappa$, weighting the NKPC (22) by α_i and summing gives (51), using $\sum_i \alpha_i n_{it} = 0$. Subtracting yields the relative NKPC (53), which depends only on $\boldsymbol{\chi}_t$ and $\mathbf{n}_t$. Orthogonality of the reallocation block follows from $(I - \bar{\mu}^2)\mathbf{1} = \mathbf{0}$: since $u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$, the operator $(I - \bar{\mu}^2)$ applied to $\mathbf{u}_t$ annihilates the $y_t\mathbf{1}$ component, so reallocation responds only to cross-sectoral differences. □

When all sectors share the same nominal rigidity, aggregate demand y_t leaves relative prices and reallocation incentives unchanged. This extends Corollary 1 to the dynamic setting: under

constant κ, mobility matters for the relative block but is irrelevant for the aggregate Phillips curve, and hence, monetary policy.

Within the relative block, two forces drive convergence after a sectoral shock: gradual price adjustment and worker reallocation. To characterize their interaction, we decompose $\bar{\mu}$ into orthogonal components along which both margins are self-contained.

Spectral decomposition. We diagonalize $\bar{\mu}$ to isolate orthogonal combinations of sectors, allowing us to independently characterize relative adjustment dynamics.[5] Detailed balance (Assumption 1) ensures $\bar{\mu}$ is diagonalizable, while a constant κ across sectors guarantees the complete decoupling of the relative block (53)–(56).

Lemma 6 (Symmetrization)**.** *Under detailed balance, $A\bar{\mu} = \bar{\mu}^T A$ where $A = \mathrm{diag}(\alpha_i)$. The matrix $Q = A^{1/2}\bar{\mu}\, A^{-1/2}$ is therefore symmetric, with eigenvalues $1 = \rho_1 > \rho_2 \geq \cdots \geq \rho_N > -1$ and orthonormal eigenvectors $\{s_k\}$. The coordinate transformation $x^*_{kt} = s_k^T A^{1/2}\hat{\mathbf{x}}_t$ for $x \in \{\chi, n, \hat{v}, \hat{a}\}$ decomposes the relative block into $N-1$ independent three-variable systems in $(\chi^*_{kt}, n^*_{kt}, \hat{v}^*_{kt})$, one for each* spectral component $k \geq 2$.

Proof. See Appendix C. □

Each eigenvector s_k isolates a *component* or a "reallocation corridor": a self-contained pattern of worker flows across sectors. Transforming back to the original sector basis, define $v_{k,i} \equiv \alpha_i^{-1/2} s_{k,i}$, so that $v_{k,i}$ is sector i's loading on component k and $x^*_{kt} = \sum_i \alpha_i\, v_{k,i}\, \hat{x}_{it}$. Throughout, indices i, j refer to sectors and k to spectral components.

The eigenvalue $|\rho_k|$ governs the speed of mean reversion along this corridor. The largest non-unit eigenvalue identifies the reallocation pattern along which worker flows are slowest to restore steady state.[6]

In transformed coordinates, component k satisfies:

$$\beta\, \mathbb{E}_t[\chi^*_{kt+1}] - [\kappa(1+\varphi) + \beta + 1]\, \chi^*_{kt} + \chi^*_{kt-1} - \kappa\varphi\, n^*_{kt} = -\Delta\hat{a}^*_{kt} + \beta\, \mathbb{E}_t[\Delta\hat{a}^*_{kt+1}], \tag{57}$$

$$n^*_{kt+1} = \rho_k\, n^*_{kt} + \phi\beta(1-\rho_k^2)\, \mathbb{E}_t[\hat{v}^*_{kt+1}], \tag{58}$$

$$\hat{v}^*_{kt} = -\chi^*_{kt} - n^*_{kt} + \beta\rho_k\, \mathbb{E}_t[\hat{v}^*_{kt+1}]. \tag{59}$$

The first equation (57) is a Phillips curve for component k, obtained by substituting relative prices (54) into the NKPC, and projecting onto component k. This is a second-order difference equation in χ^*_{kt} with n^*_{kt} entering as a cost-push shifter. The remaining two equations govern reallocation: workers flow to close the steady-state gap $-\chi^*_{kt} - n^*_{kt}$ at speeds dictated by ρ_k and ϕ. We build intuition from two limiting cases.

[5] This follows the spectral approach to spatial transition matrices in Kleinman et al. (2023) and to input-output networks in Liu and Tsyvinski (2024).

[6] Under the logit specification (12), $\bar{\mu}$ is strictly positive. By the Perron-Frobenius theorem, the dominant eigenvalue $\rho_1 = 1$ is unique and $\rho_N > -1$, ruling out non-decaying periodic oscillations.

Segmented markets ($\phi = 0$). Consider a permanent relative productivity shock $\hat{a}^*_{kt} = \hat{a}$ for $t \geq 0$. Without mobility, employment is fixed ($n^*_{kt} = 0$) and (57) becomes a univariate second-order difference equation in χ^*_{kt}. The stable solution is $\chi^*_{kt} = \lambda^t_{\text{seg}} \chi^*_{k0}$, where $\lambda_{\text{seg}} \in (0,1)$ solves:

$$\beta\lambda^2 - [\kappa(1+\varphi) + \beta + 1]\lambda + 1 = 0. \tag{60}$$

Evaluating at $t = 0$ gives $\chi^*_{k0} = \lambda_{\text{seg}}\hat{a}$ and impact inflation $\hat{\pi}^*_{k0} = (\lambda_{\text{seg}} - 1)\hat{a}$. Convergence to steady-state occurs at rate λ_{seg}.

Integrated markets ($\phi \to \infty$). When workers move freely, continuation values equalize, requiring $n^*_{kt} = -\chi^*_{kt}$ for $t \geq 1$ (Proposition 3). Substituting into (57), the φ component of the slope cancels—reallocation absorbs the intensive-margin pressure—and the stable solution is $\chi^*_{kt} = \lambda^t_{\text{int}} \chi^*_{k0}$, where $\lambda_{\text{int}} \in (0,1)$ solves

$$\beta\lambda^2 - (\kappa + \beta + 1)\lambda + 1 = 0. \tag{61}$$

At impact, employment has not yet adjusted ($n^*_{k0} = 0$), so the initial condition is:

$$\chi^*_{k0} = \frac{1}{1 + \beta + \kappa(1+\varphi) - \beta\lambda_{\text{int}}}\,\hat{a}. \tag{62}$$

Comparing (60) and (61), the only difference is the coefficient on λ: $\kappa(1+\varphi)$ versus κ. Since $\kappa\varphi > 0$, it follows that $\lambda_{\text{seg}} < \lambda_{\text{int}}$: relative price gaps decay *more slowly* under integrated markets.

Proposition 4 (Impact vs. Persistence Trade-off)**.** *Under a permanent relative productivity shock $\hat{a}^*_{kt} = \hat{a}$:*

(i) Segmented markets *yield larger initial relative inflation ($|\hat{\pi}^*_{k0}|$ higher) but faster subsequent decay (rate λ_{seg}).*

(ii) Integrated markets *yield smaller initial relative inflation but slower subsequent decay ($\lambda_{int} > \lambda_{seg}$).*

In both cases, total cumulative relative inflation is $\sum_{t=0}^{\infty} \hat{\pi}^*_{kt} = -\hat{a}$.

Proof sketch. See Appendix C. Without reallocation, segmented markets face unmitigated intensive-margin cost pressures ($\kappa\varphi$), driving a larger initial inflation jump but faster subsequent decay ($\lambda_{\text{seg}} < \lambda_{\text{int}}$). Conversely, integrated markets anticipate future cost relief via worker flows, which dampens impact inflation but slows convergence. Finally, cumulative inflation telescopes to $-\hat{a}$ because long-run relative prices must fully adjust to the shock regardless of mobility. □

The mechanism operates through forward-looking price setting. Under integrated markets, firms anticipate that worker inflows will progressively offset χ^*_{kt} through n^*_{kt}, compressing marginal-cost gaps at future horizons. This anticipated relief reduces the price adjustment firms choose *today*—dampening impact inflation—but simultaneously weakens the cost pressure that drives convergence, raising the decay rate from λ_{seg} to λ_{int}.

Persistence for arbitrary mobility. The two limiting cases are endpoints of a continuous family parametrized by ϕ. Conjecturing $\chi^*_{kt} = \lambda^t C$, $n^*_{kt} = \lambda^t N$, $\hat{v}^*_{kt} = \lambda^t V$ in (57)–(59) after shocks have died out, and eliminating V and N, yields the system's characteristic equation which pin down the persistence of relative inflation.

Proposition 5 (Persistence and Bottleneck Effect)**.** *For each component $k \geq 2$, the persistence λ of the dynamic system satisfies a degree-4 polynomial:*

$$\begin{aligned}\Big[(\lambda - \rho_k)(1 - \beta\rho_k\lambda) + \phi\beta(1 - \rho_k^2)\lambda\Big]\Big[\beta\lambda^2 - (\kappa(1+\varphi) + \beta + 1)\lambda + 1\Big] \\ + \kappa\varphi\phi\beta(1 - \rho_k^2)\lambda^2 = 0.\end{aligned} \tag{63}$$

This equation yields exactly two stable roots ($|\lambda| < 1$). Let $\lambda^*(\phi)$ denote the maximum absolute value of these stable roots, which governs the persistence of the system along component k. Then $\lambda^*(\phi)$ is continuous in ϕ, with limiting values:

$$\lim_{\phi \to 0} \lambda^*(\phi) = \max\{\lambda_{seg}, |\rho_k|\},$$
$$\lim_{\phi \to \infty} \lambda^*(\phi) = \lambda_{int}.$$

Proof. See Appendix C. □

The polynomial (63) clarifies the two margins of adjustment. The first bracketed term governs labor reallocation, the second recovers the segmented-market Phillips curve (60), and the additive term captures the coupling between them driven by ϕ.

The limiting values formalize the bottleneck effect. Moving from segmented to integrated markets *increases* persistence when price rigidity is the bottleneck ($\lambda_{\text{seg}} > |\rho_k|$), and *decreases* persistence when reallocation is the bottleneck ($|\rho_k| > \lambda_{\text{int}}$). When $\lambda_{\text{seg}} \leq |\rho_k| \leq \lambda_{\text{int}}$, the net effect is ambiguous. Comparing the slowest reallocation eigenvalue $\max_{k \geq 2} |\rho_k|$ to the price decay rate λ_{seg} provides a direct empirical test of whether the bottleneck for sectoral inflation persistence is price rigidity or labor reallocation.

4.4 Heterogeneous κ: The Double Trap and Demand Spillovers

When κ_i varies across sectors, the block separation of Lemma 5 breaks down: aggregate demand no longer shifts marginal cost uniformly, and the spectral decomposition of Section 4.3 no longer yields independent components. Two questions arise. First, does the pattern of worker flows *dissipate* or *propagate* sectoral wedges? If displaced workers flow into flexible-price sectors, those sectors adjust prices and misallocation dissipates. If instead workers flow into equally sticky sectors,neither side can adjust prices, and labor frictions and nominal rigidities compound—a *double trap*. Second, does aggregate demand—by moving sticky and flexible prices at different speeds—affect reallocation?

We retain the spectral basis of Section 4.3 because it continues to diagonalize the labor reallocation network. Heterogeneous κ_i couples the previously independent components, but treating stickiness dispersion as a perturbation captures its first-order effect as an *effective slope* for each component: a measure of whether the sectors along that labor reallocation path tend to have flexible prices.

Proposition 6 (Persistence and Reallocation)**.** *Write $\kappa_i = \bar{\kappa} + \varepsilon\,\Delta\kappa_i$ with $\sum_i \alpha_i \Delta\kappa_i = 0$ and $|\varepsilon|$ small.*

(i) Effective slope: For each component $k \geq 2$, define

$$\Delta\kappa_k^{eff} \;\equiv\; \sum_i \alpha_i\,\Delta\kappa_i\,v_{k,i}^2. \tag{64}$$

To first order in ε, the joint dynamics of relative prices and labor reallocation along component k behave as if governed by a uniform Phillips curve with slope $\bar{\kappa} + \varepsilon\,\Delta\kappa_k^{eff}$.

(ii) Persistence: Let λ_k^0 be the dominant stable root of the constant-$\bar{\kappa}$ characteristic polynomial (63) for component k. If λ_k^0 is simple and distinct from the roots of all other components $j \neq k$,

$$\lambda_k^*(\varepsilon) = \lambda_k^0 + \varepsilon\,\Delta\kappa_k^{eff} \cdot \Gamma_k + O(\varepsilon^2), \tag{65}$$

where $\Gamma_k \equiv -\partial_\kappa p(\lambda_k^0; \bar{\kappa}, \rho_k)/\partial_\lambda p(\lambda_k^0; \bar{\kappa}, \rho_k) < 0$ is a constant determined by the baseline solution under constant $\bar{\kappa}$. A negative $\Delta\kappa_k^{eff}$ strictly increases persistence.

Proof. Write $K = \bar{\kappa} I + \varepsilon\ \mathrm{diag}(\Delta\kappa_i)$. Projecting onto the $\bar{\mu}$-eigenbasis (Lemma 6), the perturbation shifts the diagonal entry of the component-k polynomial by $\varepsilon\,\Delta\kappa_k^{\mathrm{eff}}$, while off-diagonal couplings to other components enter only at $O(\varepsilon^2)$. The implicit function theorem applied to $p(\lambda;\, \bar{\kappa} + \varepsilon\,\Delta\kappa_k^{\mathrm{eff}},\, \rho_k) = 0$ at the simple root λ_k^0 yields (65). See Appendix C for the formal argument and explicit formulas for Γ_k. □

If the sectors along a reallocation path (eigenvector k) tend to be sticky ($\Delta\kappa_k^{\mathrm{eff}} < 0$), labor frictions and nominal rigidities compound into a *double trap*: the effective Phillips curve slope below $\bar{\kappa}$,

and pushing the decay rate λ_k^* closer to unity. This effect is symmetric: a sticky sector delays convergence whether it is shedding workers or absorbing them, explaining the square $v_{k,i}^2$ in equation (64).

Example. Consider three sectors where sectors A and B are sticky ($\kappa_A = \kappa_B < \bar{\kappa}$) and C is flexible ($\kappa_C > \bar{\kappa}$). A shock depresses productivity in A, requiring worker outflows. The persistence of this shock depends entirely on how the labor flows connect the sectors.

If the slowest reallocation component k connects A and C, sector C's price flexibility offsets A's stickiness: the effective slope is near $\bar{\kappa}$ and the shock dissipates at roughly the baseline rate. If instead the slowest component connects A and B, neither sectors can adjust prices, and the shock to A persists and the effective Phillips curve slope is closer to $\kappa_A < \bar{\kappa}$.

The key insight is that the shocked sector A is equally sticky in both cases, but its network position determines the persistence. In the second scenario, the presence of the flexible sector C provides no relief. In fact, even if workers move from A into C along a fast-mixing component, that adjustment only resolves the A–C margin. Sector C clears and drops out, leaving the economy trapped in the slow A–B corridor as if the flexible sector did not exist.

Aggregate demand spillover into reallocation. Heterogeneous κ_i creates a second channel: aggregate demand moves relative prices. With constant κ, a change in $\tilde{y}_t$ affects all sectoral Phillips curves uniformly, leaving relative prices unchanged. With heterogeneous κ_i, flexible sectors respond more than sticky ones, generating a relative inflation response that affects labor reallocation incentives.

Proposition 7 (Aggregate Demand Spillover)**.** *Suppose $\kappa_i = \bar{\kappa} + \Delta\kappa_i$ with $\sum_i \alpha_i \Delta\kappa_i = 0$. A unit change in $\tilde{y}_t$ shifts relative inflation in component $k \geq 2$ by $(\gamma + \varphi) d_k$, where*

$$d_k \;\equiv\; \sum_i \alpha_i \, \Delta\kappa_i \, v_{k,i} \;= \mathrm{Cov}_\alpha(\Delta\kappa_i, v_{k,i}). \tag{66}$$

A persistent change in $\tilde{y}_t$ shifts χ_{kt}^* and $n_{k,t+1}^*$ proportionally to d_k. Under constant κ, $d_k = 0$ for all $k \geq 2$, recovering block separation.

Proof. Project the demeaned Phillips curve response $MK(\gamma + \varphi)\mathbf{1}\,\tilde{y}_t$ onto the $\bar{\mu}$-eigenbasis (Lemma 6). The k-th component receives $v_k^\top A\, MK\, \mathbf{1}\,(\gamma + \varphi)\,\tilde{y}_t = (\gamma + \varphi) d_k\, \tilde{y}_t$, using $\sum_i \alpha_i v_{k,i} = 0$ for $k \geq 2$. □

The spillover coefficient d_k is exact and holds regardless of the magnitude of dispersion in κ_i. It measures the expenditure-weighted covariance between a sector's price flexibility and its *signed* loading on component k. The spillover is large when sticky and flexible sectors sit on opposite sides of a reallocation corridor: an aggregate demand expansion moves their prices at

different speeds, generating a relative price distortion χ^*_{kt} that tilts reallocation incentives along that corridor.

For example, suppose component k captures mobility between energy (flexible, loading positively) and services (sticky, loading negatively). Then $d_k > 0$: expansionary monetary policy raises energy prices faster than service prices. This relative price movement increases the marginal revenue product of labor in the energy sector, drawing workers out of sticky services and into flexible energy.

Two statistics, two channels. The effective rigidity $\Delta\kappa_k^{\text{eff}}$ and the spillover coefficient d_k capture distinct channels in each component k. The effective rigidity weights κ by the squared loading $v_{k,i}^2$, measuring whether sectors involved tend to be sticky, and governs the persistence of shocks. The spillover weights by the signed loading $v_{k,i}$, measuring whether sectors opposite sides of a corridor differ in stickiness, and governs whether aggregate demand induces cross-sector labor flows. The two statistics are independent: in a symmetric two-sector economy ($\alpha_1 = \alpha_2 = \frac{1}{2}$) with one flexible and one sticky sector ($\kappa_1 \neq \kappa_2$). The sole relative eigenvector is $v_2 = (1, -1)^\top$. Because the sectors have different stickiness, aggregate demand heavily spills over into the relative block ($d_2 = \frac{1}{2}\Delta\kappa_1 - \frac{1}{2}\Delta\kappa_2 \neq 0$): a positive output gap raises the flexible sector's relative price and actively draws workers toward it. Yet, the effective slope is exactly zero ($\Delta\kappa_2^{\text{eff}} = \frac{1}{2}\Delta\kappa_1 + \frac{1}{2}\Delta\kappa_2 = 0$).

Observability and Quantitative Validation. Both statistics are directly computable from the labor transition matrix $\bar{\mu}$ and the sectoral price adjustment frequencies. The effective rigidity (Proposition 6) are derived as first-order perturbations around uniform κ, but empirical dispersion in κ_i is large. Whether the approximation remains accurate—in particular, whether it correctly identifies which reallocation corridors are most persistent—is a quantitative question.

Section 5 addresses this by computing the exact eigenvalues of the fully coupled system from empirical transition matrices and observed price adjustment frequencies, comparing them to the first-order approximation $\lambda_k^*(\varepsilon)$.[7] If the approximation is accurate, the two statistics provide a practical diagnostic: they can be computed directly from observables and used to assess when the reallocation margin is quantitatively relevant for inflation persistence and monetary policy, which we study next.

4.5 Optimal Monetary Policy with Commitment

We now formalize the optimal monetary policy under commitment in this dynamic economy. We adopt the timeless perspective of Woodford (2003), requiring that the planner's targeting rule

[7] This part is under construction.

hold at every date. As in the static case, the efficient steady-state subsidy eliminates first-order terms, so the Ramsey problem admits a linear-quadratic approximation.

Proposition 8 (Ramsey Problem under Commitment)**.** *Under commitment in the timeless perspective, the central bank chooses a sequence* $\{\tilde{y}_t, \boldsymbol{\pi}_t, \mathbf{n}_{t+1}, \boldsymbol{\chi}_t\}_{t=0}^{\infty}$ *to minimize the intertemporal welfare loss*

$$\min \quad \mathcal{W} = \sum_{t=0}^{\infty} \beta^t \, \mathcal{L}_t, \tag{67}$$

where the per-period loss is

$$\mathcal{L}_t \overset{(2)}{=} \frac{\gamma+\varphi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1+\varphi}{2} \chi_{it}^2 + \varphi \, \chi_{it} \, n_{it} + \frac{\varphi}{2} n_{it}^2 \right], \tag{68}$$

subject to, for each sector i and date t:

$$\pi_{it} = \kappa_i [(\gamma+\varphi) \, \tilde{y}_t - (1+\varphi) \, \chi_{it} - \varphi \, n_{it}] + \beta \, \mathbb{E}_t \pi_{it+1}, \tag{69}$$

$$\chi_{it} = \chi_{it-1} + \pi_{it} - \pi_t^C + \Delta \hat{a}_{it}, \tag{70}$$

$$\mathbf{n}_{t+1} = \bar{\mu} \, \mathbf{n}_t + \phi\beta(I - \bar{\mu}^2) \, \mathbb{E}_t[\hat{\mathbf{v}}_{t+1}], \tag{71}$$

$$\hat{\mathbf{v}}_t = -\boldsymbol{\chi}_t - \mathbf{n}_t + \beta\bar{\mu} \, \mathbb{E}_t[\hat{\mathbf{v}}_{t+1}], \tag{72}$$

taking as given initial conditions $(\hat{\mathbf{p}}_{-1}, \mathbf{n}_0)$ *and the exogenous productivity process* $\{\mathbf{a}_t\}$.

Proof. The per-period loss (68) follows from a second-order expansion of household welfare around the efficient steady state (Appendix B). The constraints restate the NKPC (22), relative price identity, and reallocation block from Section 2.5. □

The Ramsey problem differs from its static analogue (Lemma 1) in two ways. First, the state vector now includes predetermined employment $\mathbf{n}_t$ alongside relative prices $\hat{\mathbf{p}}_t$, so the central bank inherits both relative price misalignment and employment misallocation from the past. Second, the reallocation block (71)–(72) creates a forward-looking channel absent in the static case: today's output gap shifts relative prices, which change reallocation incentives and thereby reshape tomorrow's employment distribution.

This channel raises the welfare cost of ignoring the reallocation margin: in the Phillips curve (69), inherited misallocation ($-\varphi n_{it}$) acts as a persistent cost-push term, and in the loss function, the terms involving n_{it} decay at a rate governed by $\bar{\mu}_i$, compounding current relative price distortions into a persistent stream of welfare losses.

First-order conditions. To state the first-order conditions compactly, define the diagonal matrices $A \equiv \text{diag}(\alpha_i)$, $K \equiv \text{diag}(\kappa_i)$, and $D_\pi \equiv \text{diag}(\alpha_i \epsilon_i / \kappa_i)$, together with $M \equiv I - \mathbf{1}\boldsymbol{\alpha}^\top$, which removes the aggregate component. Denote by $\boldsymbol{\lambda}_t$, $\boldsymbol{\eta}_t$, $\boldsymbol{\psi}_t$, and $\boldsymbol{\xi}_t$ the multipliers on

the Phillips curve, relative price identity, employment law of motion, and continuation value equation, respectively. The multiplier λ_{it} is the shadow cost of the Phillips curve in sector i. The multiplier η_{it} is the shadow value of raising sector i's relative price, and is positive when the planner would gain from a higher relative price there, negative otherwise. The multiplier ψ_{it} is the shadow cost of having one additional worker locked into sector i. The multiplier ξ_{it} is the shadow value of raising workers' continuation value in sector i, which strengthens the incentive to move into that sector.

The first-order conditions are:

$$\tilde{y}_t = \mathbf{1}^\top K \boldsymbol{\lambda}_t, \tag{F$_{\tilde{y}}$}$$

$$D_\pi \boldsymbol{\pi}_t + \boldsymbol{\lambda}_t - \boldsymbol{\lambda}_{t-1} - M^\top \boldsymbol{\eta}_t = \mathbf{0}, \tag{F$_\pi$}$$

$$A[(1+\varphi)\boldsymbol{\chi}_t + \varphi \mathbf{n}_t] + (1+\varphi) K \boldsymbol{\lambda}_t + \boldsymbol{\eta}_t - \beta \mathbb{E}_t \boldsymbol{\eta}_{t+1} + \boldsymbol{\xi}_t = \mathbf{0}, \tag{F$_\chi$}$$

$$A\varphi(\boldsymbol{\chi}_t + \mathbf{n}_t) + \varphi K \boldsymbol{\lambda}_t + \boldsymbol{\xi}_t - \bar{\mu}^\top \boldsymbol{\psi}_t + \beta^{-1} \boldsymbol{\psi}_{t-1} = \mathbf{0}, \tag{F$_n$}$$

$$\boldsymbol{\xi}_t - \bar{\mu}^\top \boldsymbol{\xi}_{t-1} - \phi\,(I - \bar{\mu}^2)^\top \boldsymbol{\psi}_{t-1} = \mathbf{0}. \tag{F$_v$}$$

Under the timeless perspective, the planner commits to the policy rule that would have been optimal had it been adopted far in the past, and the lagged multipliers $(\boldsymbol{\lambda}_{-1}, \boldsymbol{\psi}_{-1}, \boldsymbol{\xi}_{-1}$ are initialized to satisfy the stationary targeting rule.

The targeting criterion. These conditions collapse into a single targeting criterion:

Proposition 9 (Monetary Policy Target)**.** *Under commitment, optimal monetary policy is characterized by the targeting rule*

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \, \pi_{it} \;=\; (MK\mathbf{1})^\top \boldsymbol{\eta}_t = \sum_i (\kappa_i - \bar{\kappa}) \eta_{it} \tag{73}$$

where the multiplier $\boldsymbol{\eta}_t$ is determined by the system (F$_\chi$)–(F$_v$)*.*

Proof. Multiply (F$_\pi$) by $\mathbf{1}^\top K$ and use $\mathbf{1}^\top K \boldsymbol{\lambda}_t = \tilde{y}_t$ from (F$_{\tilde{y}}$). The left-hand side becomes $\mathbf{1}^\top K D_\pi \boldsymbol{\pi}_t + \tilde{y}_t - \tilde{y}_{t-1}$. Since $\mathbf{1}^\top K D_\pi = (\alpha_1\epsilon_1, \ldots, \alpha_N \epsilon_N)$, this yields the left-hand side of (73). The right-hand side follows from $\mathbf{1}^\top K M^\top = (MK\mathbf{1})^\top$. □

The left-hand side of (73) is the standard New Keynesian targeting criterion. The right-hand side is a *reallocation wedge*: it is positive when sectors that need higher relative prices to attract workers also happen to be the price-flexible sectors, and negative in the reverse case. When all sectors share the same Phillips curve slope, the wedge is zero and the standard rule applies.

When is the standard rule enough? If $\kappa_i = \kappa$ for all i, the right-hand side of (73) vanishes, and the criterion reduces to

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \, \pi_{it} = 0. \tag{74}$$

If additionally $\epsilon_i = \epsilon$, this is the textbook history-dependent rule $\tilde{y}_t - \tilde{y}_{t-1} + \epsilon \, \pi_t^C = 0$. The reason is that uniform stickiness means aggregate demand moves all sectoral prices in the same proportion, so the output gap cannot shift relative prices, and the central bank has no lever over labor reallocation. This is the dynamic counterpart to the static non-intervention result (Corollary 1).

When and in which direction does the central bank deviate? When κ_i varies across sectors, the output gap becomes a tool for shifting relative prices. A uniform increase in aggregate demand raises prices more in flexible-price sectors than in sticky-price sectors. The central bank can then use aggregate demand to generate the relative-price movements that help implement efficient reallocation.

The direction of the deviation is determined by where workers need to leave relative to where price flexibility lies. If workers need to move toward the *flexible* sectors, then they must leave the *sticky* sectors, so $\eta_{it} > 0$ where $\kappa_i < \bar{\kappa}$. The wedge is then negative and the planner contracts. Contraction lowers prices more in flexible-price sectors than in sticky-price sectors, raising the relative prices of the sticky sectors and inducing labor outflows from them. Conversely, if workers need to move toward the *sticky* sectors, then they must leave the *flexible* sectors, so $\eta_{it} > 0$ where $\kappa_i > \bar{\kappa}$. The wedge is positive and the planner stimulates. Stimulus raises prices more in the flexible sectors, lowering relative demand there and inducing labor outflows toward the sticky sectors. When the correlation between $(\kappa_i - \bar{\kappa})$ and η_{it} is zero—that is, when price flexibility and reallocation needs are unrelated—the standard criterion (74) applies.

The output gap as a tool. The aggregation result from the static case (Lemma 3) extends to the dynamic economy. Summing the NKPC (22) with weights α_i / κ_i and using $\sum_i \alpha_i n_{it} = 0$:

$$\pi_t^R = (\gamma + \varphi) \, \tilde{y}_t + \beta \, \mathbb{E}_t \pi_{t+1}^R, \tag{75}$$

independent of $\mathbf{n}_t$, $\hat{\mathbf{p}}_t$, or $\bar{\mu}$. Stabilizing $\pi_t^R = 0$ therefore closes the output gap at all dates. Gap-closing policy is nonetheless suboptimal: the targeting criterion (73) shows that the planner accepts output gap costs whenever the marginal benefit of tilting relative prices to improve labor allocation—the reallocation wedge $(MK\mathbf{1})^\top \boldsymbol{\eta}_t$—is nonzero.

How large is the deviation? The reallocation wedge depends on the shadow value η_{it}, which measures how much welfare is affected from a marginal increase in sector i's relative price. Rearranging (F_χ):

$$\boldsymbol{\eta}_t \;=\; \beta\,\mathbb{E}_t\boldsymbol{\eta}_{t+1} \;-\; \mathbf{g}_t, \qquad \mathbf{g}_t \;\equiv\; A[(1+\varphi)\,\boldsymbol{\chi}_t + \varphi\,\mathbf{n}_t] + (1+\varphi)\,K\,\boldsymbol{\lambda}_t + \boldsymbol{\xi}_t. \tag{76}$$

where $\mathbf{g}_t$ is the marginal welfare cost of the current relative-price configuration: it collects the direct misallocation loss ($A[\cdot]$), the cost-push pressure on sectoral Phillips curves ($(1+\varphi)K\boldsymbol{\lambda}_t$), and the distortion of reallocation incentives ($\boldsymbol{\xi}_t$). Forward iteration gives

$$\boldsymbol{\eta}_t \;=\; -\sum_{s=0}^{\infty}\beta^s\,\mathbb{E}_t\,\mathbf{g}_{t+s}. \tag{77}$$

Thus η_{it} equals the negative of the discounted present value of all future marginal misallocation costs associated with a marginally higher relative price in sector i. The slower $\mathbf{g}_{t+s}$ decays (more persistent shocks), the larger is $|\boldsymbol{\eta}_t|$, and the larger is the reallocation wedge on the right-hand side of (73). Persistent misallocation therefore causes optimal policy to deviate more from the standard targeting criterion (74).

Which corridors matter for policy? The targeting criterion (73) aggregates across all $N-1$ reallocation corridors. To determine which corridors drive the policy wedge, we project (73) onto the $\bar{\mu}$-eigenbasis (Lemma 6). Since $MK\mathbf{1} = K\mathbf{1} - \bar{\kappa}\mathbf{1}$ has zero α-mean, it lies in the relative subspace and decomposes as

$$MK\mathbf{1} \;=\; \sum_{k=2}^{N} v_k\,d_k, \qquad d_k \;\equiv\; v_k^\top A\,MK\mathbf{1} \;=\; \sum_i \alpha_i\,(\kappa_i - \bar{\kappa})\,v_{k,i}, \tag{78}$$

where d_k is the demand spillover coefficient from Proposition 7. Substituting into (73), we have:

Corollary 2 (Spectral Targeting Rule)**.** *Under commitment, the targeting criterion* (73) *decomposes as*

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i\epsilon_i\,\pi_{it} \;=\; \sum_{k=2}^{N} d_k\,\eta_{kt}^*, \qquad \eta_{kt}^* \equiv v_k^\top\boldsymbol{\eta}_t, \tag{79}$$

where $\eta_{kt}^* \equiv v_k^\top\boldsymbol{\eta}_t$ is the shadow value of tilting relative-price gaps in the direction of corridor k. Each corridor contributes a wedge equal to its demand spillover d_k times that shadow value.

Proof. Multiply (F_π) by $\mathbf{1}^\top K$, use $\mathbf{1}^\top K\boldsymbol{\lambda}_t = \tilde{y}_t$ from ($F_{\tilde{y}}$), and expand $MK\mathbf{1}$ in the eigenbasis. □

The corridor-level decomposition makes precise the two conditions for the policy wedge to be large along a given corridor. Projecting the Phillips curve (69) onto corridor k yields the corridor

Phillips curve:

$$\pi^*_{kt} - \beta\,\mathbb{E}_t \pi^*_{k,t+1} \;=\; (\gamma+\varphi)\, d_k\, \tilde{y}_t \;-\; (1+\varphi)\sum_j \mathcal{K}_{kj}\, \chi^*_{jt} \;-\; \varphi \sum_j \mathcal{K}_{kj}\, n^*_{jt}, \tag{80}$$

where $\pi^*_{kt} \equiv v_k^\top A \boldsymbol{\pi}_t$ is the inflation rate along corridor k (likewise for χ^*_{kt} and n^*_{kt}), and $\mathcal{K}_{kj} \equiv \sum_i \alpha_i\, \kappa_i\, v_{k,i}\, v_{j,i}$. The two network statistics from Section 4.4 appear as entries of this matrix $\mathcal{K}$:

$$d_k \;=\; \mathcal{K}_{k1}, \qquad\qquad \bar{\kappa} + \Delta\kappa_k^{\text{eff}} \;=\; \mathcal{K}_{kk}. \tag{81}$$

These two statistics govern different margins of the policy problem. The coefficient d_k governs *direct transmission*: it measures the immediate tilt in relative prices along corridor k generated by a unit increase in the output gap. When $|d_k|$ is large, aggregate demand is a powerful instrument for moving prices along that corridor; when $d_k \approx 0$, policy has little direct leverage on it, although indirect effects may still operate through the off-diagonal terms $\mathcal{K}_{kj}$.

The coefficient $\bar{\kappa} + \Delta\kappa_k^{\text{eff}}$ governs *local persistence*: a more negative $\Delta\kappa_k^{\text{eff}}$ makes the corridor effectively stickier, so misalignment corrects more slowly on its own. By the present-value representation (77), slower self-correction raises $|\eta^*_{kt}|$: the misallocation costs $\mathbf{g}_{t+s}$ decay more slowly, increasing the shadow value of tilting relative prices along that corridor.

The planner therefore deviates most from the standard targeting criterion (74) along corridors for which both conditions hold: $|d_k|$ is large, so aggregate demand can materially tilt relative prices along the corridor, and $\Delta\kappa_k^{\text{eff}}$ is negative, so the welfare costs of misalignment are persistent. When $d_k \approx 0$ for all k, aggregate demand has little direct leverage on relative prices along any corridor, the policy wedge becomes small, and the one-sector Ramsey problem (82)

$$\min_{\{\tilde{y}_t\}} \sum_{t=0}^{\infty} \beta^t \left[\frac{\gamma+\varphi}{2}\, \tilde{y}_t^2 + \frac{\epsilon}{2\kappa}\, (\pi_t^C)^2 \right] \quad \text{s.t.} \quad \pi_t^C = \kappa(\gamma+\varphi)\, \tilde{y}_t + \beta\,\mathbb{E}_t \pi_{t+1}^C + u_t \tag{82}$$

serves as a close approximation.

Conversely, when $d_k \neq 0$, the sign of the wedge contributed by corridor k is $d_k \eta^*_{kt}$, which is invariant to the arbitrary sign normalization of v_k. If $d_k \eta^*_{kt} > 0$, expansion tilts relative prices in the direction desired by the planner, and the planner runs a positive output gap. Economically, this is the case in which the side of the corridor whose prices are more responsive to aggregate demand is also the side that should shed workers: stimulus raises its relative price, lowers relative demand there, and induces labor outflows. If $d_k \eta^*_{kt} < 0$, expansion tilts relative prices in the wrong direction, and the planner contracts. In the case in which the side that should shed workers is relatively sticky, tighter policy lowers prices more on the flexible side, thereby raising the sticky side's relative price and encouraging labor to leave it.

5 Quantitative Illustration

[This section is under construction.]

We illustrate the model's key mechanisms in a two-sector economy.

5.1 Calibration

Table 2 shows our calibration. The sectors are symmetric in size but differ in price rigidity ($\kappa_1 = 0.02, \kappa_2 = 0.05$.)

Parameter	Value	Description
β_i	[0.5000, 0.5000]	Expenditure shares
θ	0.1000	Migration elasticity
β	0.9900	Discount factor
γ	1.0000	Risk aversion
φ	2.0000	Frisch elasticity
ϕ_π	1.5000	Taylor rule coefficient on inflation
κ_i	[0.0200, 0.0500]	Slope of Phillips curve

Table 2: Parameter values. Two symmetric sectors that differ in price rigidity

5.2 Impulse Responses

Figure 1 plots impulse responses to a permanent relative productivity shock under varying degrees of labor mobility. Higher mobility dampens the initial inflation response but raises persistence, consistent with Proposition 4: forward-looking firms discount future cost pressures when they anticipate worker inflows.

6 Conclusion

We have developed a multi-sector New Keynesian model in which frictional labor reallocation, summarized by a mobility elasticity and a steady-state transition matrix, reshapes the transmission of sectoral shocks and the design of optimal monetary policy. Three results stand out. First, labor mobility creates countervailing effects on sectoral inflation: it dampens impact but slows convergence. Second, the persistence of shocks depends not on average mobility or average stickiness alone, but on whether displaced workers flow into flexible or sticky sectors. Third, optimal policy under commitment deviates from standard sticky-price targeting precisely when price rigidity aligns with the slowest-mixing corridors of the reallocation network, as the planner exploits flexible sectors' price responsiveness to redirect worker flows.

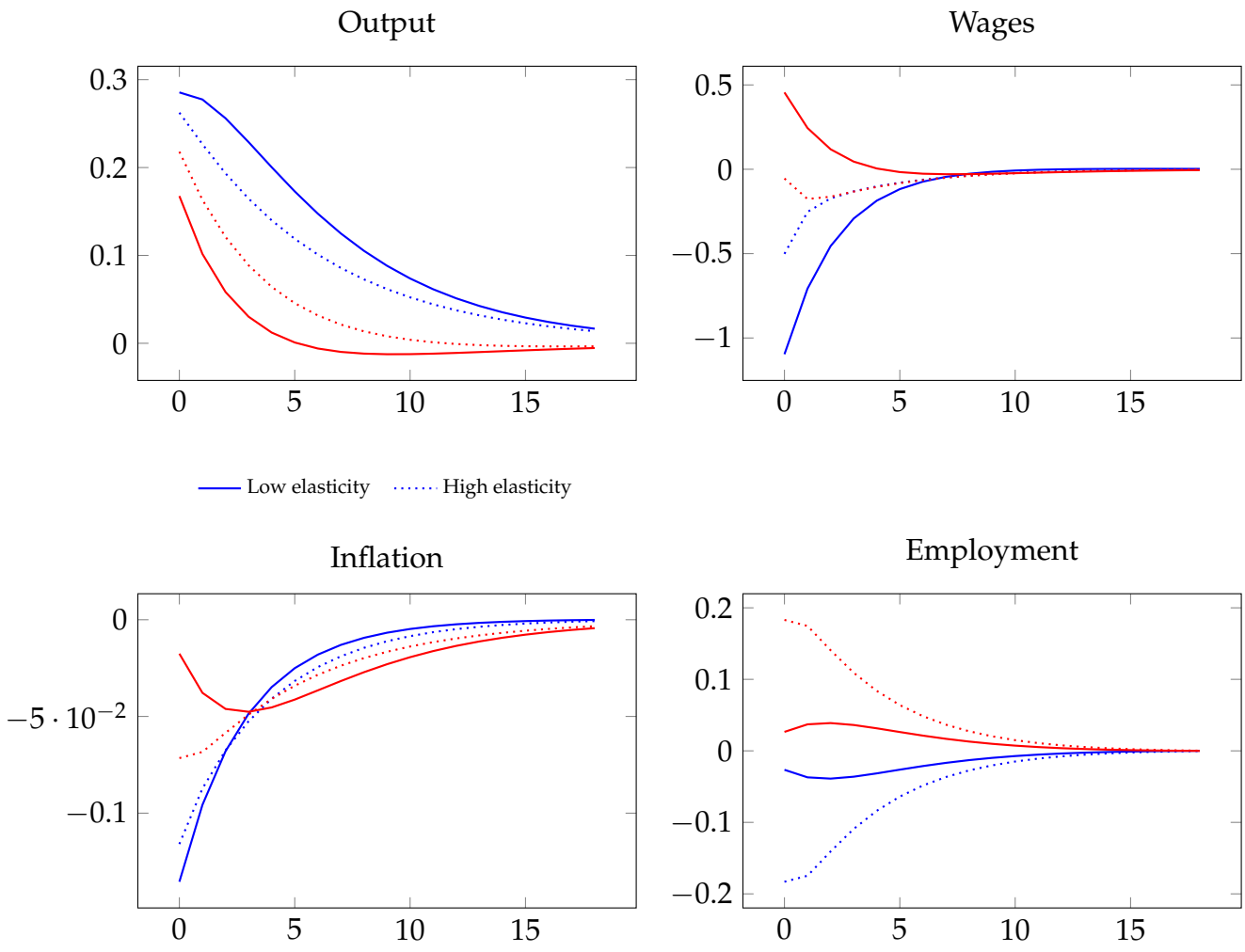

Figure 1: Response to a positive productivity shock at sector 1. Solid lines represent a model with low mobility (low θ), and dotted lines represent a sector with high mobility (low θ)

Several extensions are natural. Our model assumes that aggregate demand affects demand for all sectors symmetrically: in practice, interest rate changes have heterogeneous incidence across sectors—durables respond more than services, for instance—which would give the central bank a richer set of relative price levers. Incorporating production networks alongside the labor network would allow both input-output and worker-flow linkages to shape persistence and policy simultaneously.

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Appendix

A Log-Linearization Derivations

This appendix details the log-linearization of the model around the steady state. Because all output is consumed, we impose goods market clearing $y_t = c_t$ and $y_{it} = c_{it}$ throughout, using y to denote both output and consumption.

Notation. For any variable X_t, we write $x_t \equiv \log X_t - \log \bar{X}$ for its log-deviation from steady state. We use the following conventions throughout:

- *Natural (flexible-price) levels*: x_t^{nat} denotes the value under flexible prices.
- *Gaps*: $\tilde{x}_t \equiv x_t - x_t^{\text{nat}}$ denotes the deviation from the natural level.
- *Relative prices*: $\hat{p}_{it} \equiv p_{it} - p_t^C$ is the log sectoral price relative to the CPI.
- *Relative price distortion*: $\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}$, the deviation of relative prices from their static natural benchmark. See Section 2.5 for a discussion of how this relates to the dynamic natural relative price when $n_{it} \neq 0$.

We write $\overset{(1)}{=}$ for equalities that hold up to a first-order approximation, and $\overset{(2)}{=}$ for second-order. Plain $=$ denotes exact relationships or definitions.

A.1 The Macro Block

A.1.1 Sectoral Marginal Cost

Real marginal cost in sector i (in log-deviations) is given by $mc_{it} - p_{it} = w_{it} - a_{it} - p_{it}$. Using the intratemporal labor supply condition (8) and substituting aggregate consumption with output ($c_t = y_t$), the sectoral wage satisfies:

$$w_{it} - p_t^C = \varphi l_{it} + \gamma y_t. \tag{A.1}$$

Substituting this into the marginal cost expression yields:

$$mc_{it} - p_{it} = \gamma y_t + \varphi l_{it} - a_{it} - \hat{p}_{it},$$

where $\hat{p}_{it} \equiv p_{it} - p_t^C$ is the relative sectoral price. To eliminate l_{it}, we use the log-linearized production function ($l_{it} = y_{it} - a_{it} - n_{it}$) and substitute out sectoral output y_{it} using the demand

curve ($y_{it} = y_t - \hat{p}_{it}$). Grouping terms directly yields Equation (22) from the main text:

$$mc_{it} - p_{it} \overset{(1)}{=} (\gamma + \varphi) y_t - (1 + \varphi)(\hat{p}_{it} + a_{it}) - \varphi n_{it}. \tag{A.2}$$

A.1.2 Phillips Curve

Standard log-linearization of the Calvo pricing problem (5) yields the sectoral New Keynesian Phillips curve. A re-optimizing firm in sector i sets its price as a markup over a weighted average of current and expected future marginal costs. Log-linearizing the optimal reset price around the zero-inflation steady state, and combining with the Calvo price index evolution $P_{it} = [\delta_i P_{it-1}^{1-\epsilon_i} + (1 - \delta_i)(P_{it}^*)^{1-\epsilon_i}]^{1/(1-\epsilon_i)}$, yields:

$$\pi_{it} \overset{(1)}{=} \kappa_i (mc_{it} - p_{it}) + \beta \, \mathbb{E}_t \pi_{it+1}, \tag{A.3}$$

where $\kappa_i \equiv (1 - \delta_i)(1 - \beta\delta_i)/\delta_i$. This is Equation (21).

A.1.3 The IS Curve and Natural Real Rate

Note on convention: As is standard, we make an exception to the log-deviation convention for interest rates. We let i_t and r_t^{nat} denote the levels of the net nominal and natural real interest rates, respectively.

The Euler equation (9) is

$$1 = \beta(1 + i_t) \, \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{P_t^C}{P_{t+1}^C} \right].$$

Taking logs and using the first-order approximation $\log(1 + i_t) \overset{(1)}{=} i_t$:

$$0 \overset{(1)}{=} \log \beta + i_t - \gamma \, \mathbb{E}_t (y_{t+1} - y_t) - \mathbb{E}_t \pi_{t+1}^C,$$

where we have substituted $c_t = y_t$ (goods market clearing) and $\pi_{t+1}^C = p_{t+1}^C - p_t^C$. Rearranging yields:

$$y_t \overset{(1)}{=} \mathbb{E}_t y_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}_t \pi_{t+1}^C + \log \beta). \tag{A.4}$$

The same equation holds under flexible prices with y_t^{nat} in place of y_t and i_t^{nat} in place of i_t (and $\pi_t^C = 0$ under flexible prices). We define the natural real rate (in levels) as:

$$r_t^{\text{nat}} \equiv -\log \beta + \gamma (\mathbb{E}_t y_{t+1}^{\text{nat}} - y_t^{\text{nat}}).$$

Subtracting the flexible-price Euler equation from (A.4) gives the IS curve, Equation (23):

$$\tilde{y}_t \overset{(1)}{=} \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t \pi^C_{t+1} - r^{\text{nat}}_t\right). \tag{A.5}$$

A.1.4 Goods Market Clearing and Relative Prices.

Cobb-Douglas demand gives $P_{it}C_{it} = \alpha_i P^C_t C_t$, so sectoral output satisfies $P_{it}Y_{it} = \alpha_i P^C_t C_t$. In logs:

$$y_{it} - y_t = -\hat{p}_{it}. \tag{A.6}$$

Log aggregate output is $y_t = \sum_i \alpha_i a_{it} + \sum_i \alpha_i (n_{it} + l_{it})$, which uses $y_{it} = a_{it} + n_{it} + l_{it}$ (from the production function and labor market clearing) weighted by consumption shares.

The CPI definition $P^C_t = \prod_i P^{\alpha_i}_{it}$ implies $p^C_t = \sum_i \alpha_i p_{it}$, so CPI inflation is $\pi^C_t = \sum_i \alpha_i \pi_{it}$. Relative prices evolve as:

$$\hat{p}_{it} - \hat{p}_{it-1} = (p_{it} - p_{it-1}) - (p^C_t - p^C_{t-1}) = \pi_{it} - \pi^C_t, \tag{A.7}$$

which is Equation (24).

A.2 Dynamic Labor Reallocation Block

The dynamic reallocation block consists of the flow utility, the Bellman equation, and the employment evolution equation.

A.2.1 Flow Utility

Using the intratemporal labor supply condition (8), flow utility from (14) simplifies to $U^{\text{flow}}_{it} = \zeta \frac{\varphi}{1+\varphi} L^{1+\varphi}_{it}$ (Equation (15)). Define the normalized deviation $u_{it} \equiv (U^{\text{flow}}_{it} - \bar{U}^{\text{flow}})/(\varphi\zeta\bar{L}^{1+\varphi})$. Since $U^{\text{flow}}_{it} = \zeta \frac{\varphi}{1+\varphi} L^{1+\varphi}_{it}$, we compute:

$$\frac{U^{\text{flow}}_{it} - \bar{U}^{\text{flow}}}{\varphi\zeta\bar{L}^{1+\varphi}} = \frac{\zeta \frac{\varphi}{1+\varphi}(L^{1+\varphi}_{it} - \bar{L}^{1+\varphi})}{\varphi\zeta\bar{L}^{1+\varphi}} = \frac{L^{1+\varphi}_{it} - \bar{L}^{1+\varphi}}{(1+\varphi)\bar{L}^{1+\varphi}}.$$

A first-order expansion of $L^{1+\varphi}_{it}$ around $\bar{L}$ gives $L^{1+\varphi}_{it} \approx \bar{L}^{1+\varphi}(1 + (1+\varphi)l_{it})$, so:

$$u_{it} \overset{(1)}{=} l_{it}.$$

Substituting $l_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$ (from goods market clearing and the production function) yields Equation (27):

$$u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}. \tag{A.8}$$

A.2.2 Value Function Recursion

The Bellman equation (13) is:

$$V_{it} = U_{it}^{\text{flow}} + \frac{1}{\theta} \log\left(\sum_{j \in \mathcal{N}} \exp\left(\theta[\beta\, \mathbb{E}_t V_{jt+1} - \tau^{ij}]\right) \right).$$

In steady state, $\bar{V}_i = \bar{U}^{\text{flow}} + \frac{1}{\theta} \log\left(\sum_j \exp(\theta[\beta \bar{V}_j - \tau^{ij}])\right)$. Define the log-sum-exp term as $\Lambda_i(\mathbf{z}) \equiv \frac{1}{\theta} \log\left(\sum_j \exp(\theta z_j)\right)$ evaluated at $z_j = \beta\, \mathbb{E}_t V_{jt+1} - \tau^{ij}$. The first-order Taylor expansion of Λ_i around the steady-state values $\bar{z}_j = \beta \bar{V}_j - \tau^{ij}$ uses the standard logit property:

$$\left.\frac{\partial \Lambda_i}{\partial z_j}\right|_{\bar{z}} = \frac{\exp(\theta \bar{z}_j)}{\sum_{j'} \exp(\theta \bar{z}_{j'})} = \bar{\mu}^{ij},$$

where $\bar{\mu}^{ij}$ is the steady-state transition probability from sector i to sector j (Equation (12) evaluated at steady state). Therefore:

$$\Lambda_i(\mathbf{z}_t) - \Lambda_i(\bar{\mathbf{z}}) \overset{(1)}{=} \sum_j \bar{\mu}^{ij}\, \beta(\mathbb{E}_t V_{jt+1} - \bar{V}_j).$$

Subtracting the steady-state Bellman from the time-t Bellman and normalizing by $\varphi \zeta \bar{L}^{1+\varphi}$:

$$v_{it} = u_{it} + \beta \sum_j \bar{\mu}^{ij}\, \mathbb{E}_t[v_{jt+1}].$$

In matrix notation, this is Equation (26):

$$\mathbf{v}_t = \mathbf{u}_t + \beta \bar{\mu}\, \mathbb{E}_t[\mathbf{v}_{t+1}]. \tag{A.9}$$

A.2.3 Employment Evolution

Employment evolves according to $N_{jt+1} = \sum_i \mu_t^{ij} N_{it}$, where the transition probabilities are given by the logit form (12):

$$\mu_t^{ij} = \frac{\exp\left(\theta[\beta\, \mathbb{E}_t V_{jt+1} - \tau^{ij}]\right)}{\sum_{j'} \exp\left(\theta[\beta\, \mathbb{E}_t V_{j't+1} - \tau^{ij'}]\right)}.$$

Linearizing μ_t^{ij}. The transition probability has the logit form, so its first-order expansion around the steady state $(\bar{\mu}^{ij}, \bar{V}_j)$ is:

$$\begin{aligned}\mu_t^{ij} - \bar{\mu}^{ij} &\overset{(1)}{=} \bar{\mu}^{ij}\,\theta\beta\left(\mathbb{E}_t[V_{jt+1} - \bar{V}_j] - \sum_{j'} \bar{\mu}^{ij'}\,\mathbb{E}_t[V_{j't+1} - \bar{V}_{j'}]\right)\\ &= \bar{\mu}^{ij}\,\theta\beta\,\varphi\zeta\bar{L}^{1+\varphi}\left(\mathbb{E}_t[v_{jt+1}] - \sum_{j'} \bar{\mu}^{ij'}\,\mathbb{E}_t[v_{j't+1}]\right),\end{aligned}$$

where the second line uses the normalization $v_{jt} = (V_{jt} - \bar{V}_j)/(\varphi\zeta\bar{L}^{1+\varphi})$ and defines $\phi \equiv \varphi\theta\zeta\bar{L}^{1+\varphi}$. Noting that $\sum_{j'} \bar{\mu}^{ij'}\mathbb{E}_t[v_{j't+1}]$ is the $\bar{\mu}$-weighted average from origin i, we have:

$$\mu_t^{ij} - \bar{\mu}^{ij} \overset{(1)}{=} \phi\beta\,\bar{\mu}^{ij}\left(\mathbb{E}_t[v_{jt+1}] - \sum_{j'} \bar{\mu}^{ij'}\,\mathbb{E}_t[v_{j't+1}]\right). \tag{A.10}$$

Deriving the employment law of motion. Log-linearize $N_{jt+1} = \sum_i \mu_t^{ij} N_{it}$ around $\bar{N}_j = \alpha_j$. Since $\bar{N}_j = \sum_i \bar{\mu}^{ij}\bar{N}_i$, taking deviations:

$$\bar{N}_j\, n_{jt+1} \overset{(1)}{=} \sum_i \bar{\mu}^{ij}\bar{N}_i\, n_{it} + \sum_i \bar{N}_i(\mu_t^{ij} - \bar{\mu}^{ij}).$$

We handle the two terms in turn. For the first term, detailed balance (16) gives $\bar{N}_i\bar{\mu}^{ij} = \bar{N}_j\bar{\mu}^{ji}$, so:

$$\sum_i \bar{\mu}^{ij}\bar{N}_i\, n_{it} = \bar{N}_j \sum_i \bar{\mu}^{ji} n_{it}.$$

Reading this entry-wise, the j-th component is $\bar{N}_j[\bar{\mu}^\top \mathbf{n}_t]_j$. However, because we work with the log-deviations $n_{it} = \log(N_{it}/\bar{N}_i)$ and impose detailed balance, the symmetrization $\bar{N}_i\bar{\mu}^{ij} = \bar{N}_j\bar{\mu}^{ji}$ allows us to write this equivalently (dividing through by $\bar{N}_j$) as:

$$\text{(first term)} = \sum_i \bar{\mu}^{ji} n_{it} = [\bar{\mu}\,\mathbf{n}_t]_j,$$

where the last equality exploits the fact that under detailed balance with symmetric steady state ($\bar{N}_i = \alpha_i$), the operator acting on $\mathbf{n}_t$ can be written as $\bar{\mu}$ without transposes (see Remark below).

For the second term, substituting (A.10):

$$\sum_i \bar{N}_i(\mu_t^{ij} - \bar{\mu}^{ij}) = \phi\beta \sum_i \bar{N}_i\bar{\mu}^{ij}\left(\mathbb{E}_t[v_{jt+1}] - \sum_{j'} \bar{\mu}^{ij'}\mathbb{E}_t[v_{j't+1}]\right).$$

Using detailed balance ($\bar{N}_i \bar{\mu}^{ij} = \bar{N}_j \bar{\mu}^{ji}$), dividing by $\bar{N}_j$, and writing the result in matrix form:

$$\begin{aligned}(\text{second term})/\bar{N}_j &= \phi\beta\left(\mathbb{E}_t[v_{jt+1}]\sum_i \bar{\mu}^{ji} - \sum_i \bar{\mu}^{ji}\sum_{j'} \bar{\mu}^{ij'}\mathbb{E}_t[v_{j't+1}]\right) \\ &= \phi\beta\left(\mathbb{E}_t[v_{jt+1}] - [\bar{\mu}^2\,\mathbb{E}_t\mathbf{v}_{t+1}]_j\right),\end{aligned}$$

where we used $\sum_i \bar{\mu}^{ji} = 1$ and the fact that $\sum_i \bar{\mu}^{ji}\sum_{j'} \bar{\mu}^{ij'}\mathbb{E}_t[v_{j't+1}]$ is the j-th entry of $\bar{\mu}^2\,\mathbb{E}_t\mathbf{v}_{t+1}$ (again using detailed balance to avoid transposes).

Combining both terms:

$$n_{jt+1} = [\bar{\mu}\,\mathbf{n}_t]_j + \phi\beta\Big[(I-\bar{\mu}^2)\,\mathbb{E}_t\mathbf{v}_{t+1}\Big]_j. \tag{A.11}$$

In vector form, this is Equation (25):

$$\mathbf{n}_{t+1} = \bar{\mu}\,\mathbf{n}_t + \phi\beta(I-\bar{\mu}^2)\,\mathbb{E}_t[\mathbf{v}_{t+1}]. \tag{A.12}$$

Remark 1 (Transpose under detailed balance)**.** *In general, $N_{jt+1} = \sum_i \mu_t^{ij} N_{it}$ would give the operator $\bar{\mu}^\top$ on $\mathbf{N}_t$ (since the sum is over the origin i). However, we work with log-deviations $n_{it} = \log(N_{it}/\bar{N}_i)$ rather than levels. Detailed balance implies $\bar{N}_i\bar{\mu}^{ij} = \bar{N}_j\bar{\mu}^{ji}$, or equivalently $A\bar{\mu} = \bar{\mu}^\top A$ where $A = \mathrm{diag}(\alpha_i)$. This conjugacy, combined with dividing the evolution equation by $\bar{N}_j$, converts $\bar{\mu}^\top$ into $\bar{\mu}$ when acting on the proportional deviations $\mathbf{n}_t$. The same symmetrization applies to the $\bar{\mu}^2$ term.*

Limiting cases. When $\phi = 0$ (segmented markets), the second term vanishes and $\mathbf{n}_{t+1} = \bar{\mu}\,\mathbf{n}_t$. Starting from $\mathbf{n}_0 = \mathbf{0}$, we have $\mathbf{n}_t = \mathbf{0}$ for all t: no reallocation occurs. When $\phi \to \infty$, boundedness of $\mathbf{n}_{t+1}$ requires $(I-\bar{\mu}^2)\,\mathbb{E}_t[\mathbf{v}_{t+1}] \to \mathbf{0}$. Since $I-\bar{\mu}^2$ has a nontrivial null space only along the constant vector $\mathbf{1}$ (which is already removed by the constraint $\sum_i \alpha_i n_{it} = 0$), this implies $\mathbb{E}_t[\mathbf{v}_{t+1}] \to \mathbf{1}c_t$. Iterating the value recursion (26) then gives $\mathbf{u}_t = \mathbf{1}c_t$, so $n_{it} = -(y_t - \hat{p}_{it} - a_{it}) = -\chi_{it}$ from (27) and $\sum_i \alpha_i n_{it} = 0$: employment fully offsets relative price distortions.

A.3 Static Economy and the Reduced-Form Sectoral NKPC

In the static limit, continuation values are common across destinations and cancel from the logit choice, so reallocation occurs within the period.

A.3.1 Static Reallocation Condition

In the static model, the dynamic transition friction is replaced by a within-period discrete choice problem. Workers choose their sector contemporaneously to maximize current flow

utility, subject to idiosyncratic preference shocks that anchor aggregate shares to the steady-state baseline $\bar{N}_i$. As established in Section 3, this yields the standard static logit choice rule for the cross-sectional employment distribution:

$$N_{it} = \frac{\bar{N}_i \exp(\theta U_{it}^{\text{flow}})}{\sum_j \bar{N}_j \exp(\theta U_{jt}^{\text{flow}})}.$$

Taking logs:

$$\log N_{it} = \log \bar{N}_i + \theta U_{it}^{\text{flow}} - \log\left(\sum_j \bar{N}_j \exp(\theta U_{jt}^{\text{flow}})\right).$$

Subtracting $\log \bar{N}_i$ and expanding the log-share expression in deviations of U^{flow}, we get:

$$n_{it} \overset{(1)}{=} \theta(U_{it}^{\text{flow}} - \bar{U}^{\text{flow}}) - \sum_j \alpha_j \theta(U_{jt}^{\text{flow}} - \bar{U}^{\text{flow}}).$$

The second term arises from log-linearizing the denominator: $\log\left(\sum_j \bar{N}_j \exp(\theta U_{jt}^{\text{flow}})\right) \overset{(1)}{=} \theta \bar{U}^{\text{flow}} + \theta \sum_j \alpha_j (U_{jt}^{\text{flow}} - \bar{U}^{\text{flow}})$. Using the result from Appendix A.2.1 that $(U_{it}^{\text{flow}} - \bar{U}^{\text{flow}})/(\varphi\zeta\bar{L}^{1+\varphi}) \overset{(1)}{=} l_{it}$, and defining $\phi \equiv \varphi\theta\zeta\bar{L}^{1+\varphi}$:

$$n_{it} = \phi(l_{it} - l_t^{\alpha}), \qquad l_t^{\alpha} \equiv \sum_i \alpha_i l_{it}, \tag{A.13}$$

which is Equation (30). The constraint $\sum_i \alpha_i n_{it} = 0$ is satisfied by construction, since $\sum_i \alpha_i (l_{it} - l_t^{\alpha}) = 0$.

A.3.2 Derivation of the Sectoral NKPC

We derive the reduced-form Phillips curve (39). Substituting the marginal cost expression into the sectoral NKPC:

$$\pi_{it} = \kappa_i \left[(\gamma + \varphi)\tilde{y}_t - \sigma\chi_{it}\right] + \beta \mathbb{E}_t \pi_{it+1}. \tag{A.14}$$

The relative price distortion evolves as $\chi_{it} = \chi_{it-1} + (\pi_{it} - \pi_t^C) + \Delta\hat{a}_{it}$. Define $\hat{c}_{it} \equiv \chi_{it-1} + \Delta\hat{a}_{it}$, which collects the inherited price misalignment and the current productivity shock. Substituting into (A.14) and collecting π_{it} on the left-hand side gives, after dividing by $(1 + \sigma\kappa_i)$ and using $\bar{\kappa}_i \equiv \sigma\kappa_i/(1 + \sigma\kappa_i)$:

$$\pi_{it} = \frac{\gamma + \varphi}{\sigma}\bar{\kappa}_i \tilde{y}_t + \bar{\kappa}_i \pi_t^C - \bar{\kappa}_i \hat{c}_{it} + \frac{\beta}{1 + \sigma\kappa_i}\mathbb{E}_t \pi_{it+1}. \tag{A.15}$$

Aggregating with weights α_i and solving for π_t^C:

$$\pi_t^C = \frac{\gamma+\varphi}{\sigma}\frac{\bar{\kappa}^\alpha}{1-\bar{\kappa}^\alpha}\tilde{y}_t - \frac{1}{1-\bar{\kappa}^\alpha}\sum_j \alpha_j\bar{\kappa}_j\hat{c}_{jt} + \frac{\beta}{1-\bar{\kappa}^\alpha}\sum_j \frac{\alpha_j}{1+\sigma\kappa_j}\mathbb{E}_t\pi_{jt+1}. \tag{A.16}$$

where $\bar{\kappa}^\alpha \equiv \sum_i \alpha_i\bar{\kappa}_i$. Substituting (A.16) back into (A.15):

$$\begin{aligned}\pi_{it} = b_i\tilde{y}_t - \frac{\bar{\kappa}_i}{1-\bar{\kappa}^\alpha}\left[(1-\bar{\kappa}^\alpha)\hat{c}_{it} + \sum_j \alpha_j\bar{\kappa}_j\hat{c}_{jt}\right] \\ + \frac{\beta}{1+\sigma\kappa_i}\mathbb{E}_t\pi_{it+1} + \frac{\beta\bar{\kappa}_i}{1-\bar{\kappa}^\alpha}\sum_j \frac{\alpha_j}{1+\sigma\kappa_j}\mathbb{E}_t\pi_{jt+1},\end{aligned} \tag{A.17}$$

where the slope coefficient is $b_i = \frac{(\gamma+\varphi)\kappa_i}{(1+\sigma\kappa_i)(1-\bar{\kappa}^\alpha)}$.

Verification: constant κ. When $\kappa_i = \kappa$ for all i, we have $\bar{\kappa}_i = \bar{\kappa} \equiv \frac{\sigma\kappa}{1+\sigma\kappa}$ and $\bar{\kappa}^\alpha = \bar{\kappa}$, so $1-\bar{\kappa}^\alpha = \frac{1}{1+\sigma\kappa}$. The output gap slope simplifies to $b_i = (\gamma+\varphi)\kappa$, independent of σ. This confirms Corollary 1: with uniform price stickiness, mobility affects the Phillips curve only through the cost-push terms and relative price dynamics, not through the output gap slope.

B Second-Order Expansion Details

This appendix derives the per-period welfare loss functions stated in Proposition 1 and 8. We expand household utility around the efficient (natural) allocation to second order, term by term.

B.1 Proposition 1: Static Loss Function

We derive Equation 42:

$$\mathcal{L}_t \overset{(2)}{=} \frac{\gamma+\varphi}{2}\tilde{y}_t^2 + \sum_i \frac{\alpha_i\epsilon_i}{2\kappa_i}\pi_{it}^2 + \frac{\sigma}{2}\sum_i \alpha_i\chi_{it}^2,$$

The per-period utility is

$$U_t = \frac{C_t^{1-\gamma}}{1-\gamma} - \sum_i N_{it}\,\zeta\frac{L_{it}^{1+\varphi}}{1+\varphi} - \Psi_t, \tag{B.1}$$

where $\Psi_t = \frac{1}{\theta}\sum_i N_{it}\log(N_{it}/\bar{N}_i)$ is the static reallocation cost from (11). We expand each of the three terms around the natural allocation $(C_t^{\text{nat}}, L_{it}^{\text{nat}}, N_{it}^{\text{nat}})$, normalizing throughout by $U_C\bar{C}^{\text{nat}} = (\bar{C}^{\text{nat}})^{1-\gamma}$. The planner first-order condition on L_{it} (Equation (32)) gives $\alpha_i(\bar{C}^{\text{nat}})^{1-\gamma} = \zeta\,\alpha_i\,\bar{L}^{1+\varphi}$, or equivalently $\zeta\bar{L}^{1+\varphi} = U_C\bar{C}^{\text{nat}}$, a normalization used repeatedly below.

B.1.1 Consumption

A standard second-order expansion of the utility gives:

$$\frac{1}{U_C \bar{C}^{\text{nat}}} \frac{C_t^{1-\gamma} - (C_t^{\text{nat}})^{1-\gamma}}{1-\gamma} \overset{(2)}{=} \tilde{y}_t + \frac{1-\gamma}{2} \tilde{y}_t^2, \tag{B.2}$$

where $\tilde{y}_t = \log(C_t / C_t^{\text{nat}})$ is the output gap (using goods market clearing $c_t = y_t$).

B.1.2 Labor Disutility

This term requires care because both N_{it} and L_{it} deviate from their natural levels simultaneously. $N_{it} = \alpha_i e^{n_{it}}$ and the adding-up constraint $\sum_i \alpha_i e^{n_{it}} = 1$ implies

$$\sum_i \alpha_i n_{it} \overset{(2)}{=} -\tfrac{1}{2} \sum_i \alpha_i \, n_{it}^2, \tag{B.3}$$

we can expand $N_{it} \overset{(2)}{=} \alpha_i (1 + n_{it} + \frac{1}{2} n_{it}^2)$. Writing $L_{it} = L_{it}^{\text{nat}} \, e^{\tilde{l}_{it}}$ and expanding $L_{it}^{1+\varphi}$ to second order, the product $N_{it} \, L_{it}^{1+\varphi}$ becomes

$$N_{it} \, L_{it}^{1+\varphi} \overset{(2)}{=} \alpha_i (L_{it}^{\text{nat}})^{1+\varphi} \left[1 + n_{it} + \tfrac{1}{2} n_{it}^2 + (1+\varphi) \, \tilde{l}_{it} + \tfrac{(1+\varphi)^2}{2} \, \tilde{l}_{it}^2 + (1+\varphi) \, \tilde{l}_{it} \, n_{it} \right].$$

Using the planner FOC to normalize, summing over sectors, and applying the second-order constraint (B.3):

$$\begin{aligned} &\frac{1}{U_C \bar{C}^{\text{nat}}} \sum_i \zeta \left[\frac{N_{it} \, L_{it}^{1+\varphi}}{1+\varphi} - \frac{\alpha_i \, \bar{L}^{1+\varphi}}{1+\varphi} \right] \\ &\overset{(2)}{=} \sum_i \frac{\alpha_i}{1+\varphi} \left[n_{it} + \tfrac{1}{2} \left(n_{it}^2 - \textstyle\sum_j \alpha_j \, n_{jt}^2 \right) + (1+\varphi) \, \tilde{l}_{it} + \tfrac{(1+\varphi)^2}{2} \, \tilde{l}_{it}^2 + (1+\varphi) \, \tilde{l}_{it} \, n_{it} \right]. \end{aligned} \tag{B.4}$$

We simplify each group of terms using the second-order constraint $\sum_i \alpha_i n_{it} \overset{(2)}{=} -\frac{1}{2} \sum_i \alpha_i n_{it}^2$:

- The n_{it} and n_{it}^2 terms perfectly cancel: $\sum_i \frac{\alpha_i}{1+\varphi} \left(n_{it} + \frac{1}{2} n_{it}^2 \right) \overset{(2)}{=} 0$.
- The $\tilde{l}_{it}$ terms: $\sum_i \alpha_i \, \tilde{l}_{it} = \tilde{l}_t^{\alpha}$.
- The $\tilde{l}_{it}^2$ terms: decompose as $\sum_i \alpha_i \, \tilde{l}_{it}^2 = (\tilde{l}_t^{\alpha})^2 + \mathcal{M}_t$, where $\mathcal{M}_t \equiv \sum_i \alpha_i (\tilde{l}_{it} - \tilde{l}_t^{\alpha})^2$ is cross-sector labor dispersion.
- The cross terms: using $n_{it} = \phi(\tilde{l}_{it} - \tilde{l}_t^{\alpha})$ from (30),

$$\sum_i \alpha_i \, \tilde{l}_{it} \, n_{it} = \sum_i \alpha_i \, n_{it} (\tilde{l}_{it} - \tilde{l}_t^{\alpha}) = \phi \sum_i \alpha_i (\tilde{l}_{it} - \tilde{l}_t^{\alpha})^2 = \phi \, \mathcal{M}_t.$$

Collecting:

$$\frac{1}{U_C \bar{C}^{\text{nat}}} \sum_i \zeta \left[\frac{N_{it}\, L_{it}^{1+\varphi}}{1+\varphi} - \frac{\alpha_i\, \bar{L}^{1+\varphi}}{1+\varphi} \right] \overset{(2)}{=} \tilde{l}_t^{\alpha} + \frac{1+\varphi}{2} \left((\tilde{l}_t^{\alpha})^2 + \mathcal{M}_t \right) + \phi\, \mathcal{M}_t. \tag{B.5}$$

B.1.3 Mobility Cost

The static reallocation cost is $\Psi_t = \frac{1}{\theta} \sum_i N_{it} \log(N_{it}/\bar{N}_i) = \frac{1}{\theta} \sum_i N_{it}\, n_{it}$. We expand $N_{it}\, n_{it}$ to second order using $N_{it} \overset{(1)}{=} \alpha_i (1 + n_{it})$:

$$\sum_i N_{it}\, n_{it} \overset{(2)}{=} \sum_i \alpha_i (1 + n_{it}) n_{it} = \sum_i \alpha_i\, n_{it} + \sum_i \alpha_i\, n_{it}^2$$

Substituting the second-order constraint $\sum_i \alpha_i n_{it} \overset{(2)}{=} -\frac{1}{2} \sum_i \alpha_i n_{it}^2$ simplifies this directly:

$$\Psi_t \overset{(2)}{=} \frac{1}{\theta} \left(-\frac{1}{2} \sum_i \alpha_i\, n_{it}^2 + \sum_i \alpha_i\, n_{it}^2 \right) = \frac{1}{2\theta} \sum_i \alpha_i\, n_{it}^2$$

Substituting $n_{it} = \phi(\tilde{l}_{it} - \tilde{l}_t^{\alpha})$ and normalizing by $U_C \bar{C}^{\text{nat}}$:

$$\frac{\Psi_t}{U_C \bar{C}^{\text{nat}}} \overset{(2)}{=} \frac{\phi^2}{2\theta\, U_C \bar{C}^{\text{nat}}}\, \mathcal{M}_t = \frac{\phi\varphi}{2}\, \mathcal{M}_t, \tag{B.6}$$

where the last step uses $\phi = \varphi\theta\zeta \bar{L}^{1+\varphi} = \varphi\theta\, U_C \bar{C}^{\text{nat}}$ (by the planner FOC), so $\phi^2 / (\theta\, U_C \bar{C}^{\text{nat}}) = \phi\varphi$.

B.1.4 Assembly and Final Loss

Combining (B.2), (B.5), and (B.6):

$$\frac{U_t - U_t^{\text{nat}}}{U_C \bar{C}^{\text{nat}}} \overset{(2)}{=} \left(\tilde{y}_t + \frac{1-\gamma}{2}\, \tilde{y}_t^2 \right) - \left(\tilde{l}_t^{\alpha} + \frac{1+\varphi}{2} (\tilde{l}_t^{\alpha})^2 + \left(\frac{1+\varphi}{2} + \phi \right) \mathcal{M}_t \right) - \frac{\phi\varphi}{2}\, \mathcal{M}_t. \tag{B.7}$$

Since $\tilde{l}_t^{\alpha} \overset{(1)}{=} \tilde{y}_t$ (Lemma 1), we can replace $(\tilde{l}_t^{\alpha})^2 \overset{(2)}{=} \tilde{y}_t^2$ and combine the squared terms:

$$\frac{U_t - U_t^{\text{nat}}}{U_C \bar{C}^{\text{nat}}} \overset{(2)}{=} (\tilde{y}_t - \tilde{l}_t^{\alpha}) - \frac{\gamma + \varphi}{2}\, \tilde{y}_t^2 - \left(\frac{1+\varphi}{2} + \phi + \frac{\phi\varphi}{2} \right) \mathcal{M}_t. \tag{B.8}$$

B.1.5 The Output–Labor Gap

The remaining first-order term $\tilde{y}_t - \tilde{l}_t^{\alpha}$ vanishes at first order but has a nonzero second-order component arising from two sources of misallocation: within-sector price dispersion and across-sector employment dispersion.

Within-sector. Define the sectoral dispersion index $\Delta_{it} \equiv \int_0^1 (P_{it}(f)/P_{it})^{-\epsilon_i}\, df \geq 1$ and $\xi_{it} \equiv -\log \Delta_{it}$. This captures misallocation across varieties within sector i: effective production satisfies $Y_{it} = A_{it}\, e^{\xi_{it}}\, N_{it}\, L_{it}$. Under Calvo pricing, price dispersion evolves recursively: to second order, its law of motion is $\xi_{it} \overset{(2)}{=} \delta_i \xi_{i,t-1} - \frac{\epsilon_i \delta_i}{2(1-\delta_i)} \pi_{it}^2$. As is standard in New Keynesian welfare derivations, assuming the economy starts from a steady state with zero price dispersion ($\xi_{i,-1} = 0$), we can rewrite the discounted sum of these dispersion terms over the infinite horizon. The present-value equivalence is:

$$\sum_{t=0}^{\infty} \beta^t \xi_{it} \overset{(2)}{=} -\sum_{t=0}^{\infty} \beta^t \frac{\epsilon_i}{2\kappa_i} \pi_{it}^2 \tag{B.9}$$

Cross Sector Dispersion. Cobb-Douglas aggregation gives $C_t = \prod_i (A_{it}\, e^{\xi_{it}}\, N_{it}\, L_{it})^{\alpha_i}$. Taking logs and expressing in gaps from the natural allocation:

$$\tilde{y}_t = \sum_i \alpha_i\, \tilde{l}_{it} + \sum_i \alpha_i\, \xi_{it} + \sum_i \alpha_i \log \frac{N_{it}}{N_{it}^{\text{nat}}} \tag{B.10}$$

This is exact. The first term is $\tilde{l}_t^{\alpha}$. For the third term, since $\log(N_{it}/N_{it}^{\text{nat}}) = n_{it}$, we can apply the second-order constraint directly:

$$\sum_i \alpha_i \log \frac{N_{it}}{N_{it}^{\text{nat}}} = \sum_i \alpha_i n_{it} \overset{(2)}{=} -\frac{1}{2} \sum_i \alpha_i n_{it}^2$$

Substituting (B.9) and $n_{it} = \phi(\tilde{l}_{it} - \tilde{l}_t^{\alpha})$:

$$\tilde{y}_t - \tilde{l}_t^{\alpha} \overset{(2)}{=} -\sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 - \frac{\phi^2}{2} \mathcal{M}_t \tag{B.11}$$

in discounted present value terms.

B.1.6 Final Loss Function

Substituting (B.11) into (B.8):

$$\mathcal{L}_t \equiv -\frac{U_t - U_t^{\text{nat}}}{U_C \bar{C}^{\text{nat}}} \overset{(2)}{=} \frac{\gamma + \varphi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 + \underbrace{\left(\frac{1+\varphi}{2} + \phi + \frac{\phi\varphi}{2} + \frac{\phi^2}{2} \right)}_{\equiv\ \Omega} \mathcal{M}_t. \tag{B.12}$$

It remains to express $\mathcal{M}_t$ in terms of relative prices and simplify Ω. From the margin

decomposition (35), $\tilde{l}_{it} - \tilde{l}_t^{\alpha} = -\frac{1}{1+\phi}\chi_{it}$, so $\mathcal{M}_t = \frac{1}{(1+\phi)^2}\sum_i \alpha_i \chi_{it}^2$. The coefficient simplifies as

$$\begin{aligned}\frac{\Omega}{(1+\phi)^2} &= \frac{1}{(1+\phi)^2}\left(\frac{1+\varphi}{2} + \phi + \frac{\phi\varphi}{2} + \frac{\phi^2}{2}\right) = \frac{1}{(1+\phi)^2}\cdot\frac{1+\varphi+2\phi+\phi\varphi+\phi^2}{2} \\ &= \frac{(1+\phi)(1+\phi+\varphi)}{2(1+\phi)^2} = \frac{1+\phi+\varphi}{2(1+\phi)} = \frac{\sigma}{2}, \end{aligned} \tag{B.13}$$

where $\sigma \equiv 1 + \frac{\varphi}{1+\phi} = \frac{1+\phi+\varphi}{1+\phi}$. Substituting, we get

$$\mathcal{L}_t \overset{(2)}{=} \frac{\gamma+\varphi}{2}\,\tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i}\,\pi_{it}^2 + \frac{\sigma}{2}\sum_i \alpha_i\,\chi_{it}^2, \tag{B.14}$$

which is Equation (42) in Proposition 1. The coefficient $\sigma \in [1, 1+\varphi)$ ranges from $1+\varphi$ (segmented markets, $\phi = 0$) to 1 (frictionless mobility, $\phi \to \infty$), confirming that σ is a sufficient statistic for mobility in the welfare loss.

Remark 2 (Limiting cases)**.** *When $\phi = 0$, the mobility cost and employment dispersion terms vanish, and $\Omega\,\mathcal{M}_t = \frac{1+\varphi}{2}\sum_i \alpha_i\,\chi_{it}^2$: all adjustment occurs through the intensive margin (hours). When $\phi \to \infty$, workers fully absorb relative price distortions ($\mathcal{M}_t \to 0$), and the residual coefficient $\sigma \to 1$ reflects only expenditure misallocation from relative price gaps, recovering the welfare loss in Rubbo (2023).*

B.2 Proposition 8: Dynamic LQ

We derive the per-period welfare loss under the timeless perspective:

$$\mathcal{L}_t \overset{(2)}{=} \frac{\gamma+\varphi}{2}\,\tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i}\,\pi_{it}^2 + \sum_i \alpha_i\left[\frac{1+\varphi}{2}\,\chi_{it}^2 + \varphi\,\chi_{it}\,n_{it} + \frac{\varphi}{2}\,n_{it}^2\right].$$

The derivation proceeds in two steps. Step 1 expands the consumption-minus-labor-disutility component $u(C_t) - \sum_i N_{it}\,v(L_{it})$ to second order in steady-state deviations. Step 2 shows that the mobility cost Ψ_t is a term whose discounted sum telescopes into a boundary term that enters the Ramsey Lagrangian only through the date-0 first-order conditions. Under the timeless perspective, the per-period loss is therefore the second-order expansion of $u - \sum Nv$ alone.

Notation. All variables denote log-deviations from the efficient steady state, as defined in Appendix A. We write $h_{it} \equiv l_{it} - l_t^{\alpha}$ for cross-sector hours and normalize by $U_C\bar{C} = \bar{C}^{1-\gamma} = \zeta\bar{L}^{1+\varphi}$. Three structural identities are used throughout (same as Appendix B.1):

(i) *Production:* $y_t = \bar{a}_t + l_t^{\alpha} + \sum_i \alpha_i\,n_{it} + \sum_i \alpha_i\,\xi_{it}$, where $\xi_{it} \equiv -\log\Delta_{it} \leq 0$ is the within-sector price-dispersion wedge. This is exact.

(ii) *Cross-section:* $h_{it} \overset{(1)}{=} -(\chi_{it} + n_{it})$, from goods market clearing and the production function.

(iii) *Adding-up:* $\sum_i \alpha_i\, n_{it} \overset{(2)}{=} -\frac{1}{2}\sum_i \alpha_i\, n_{it}^2$, from $\sum_i \alpha_i\, e^{n_{it}} = 1$.

B.2.1 Second-Order Expansion of $u(C_t) - \sum_i N_{it}\, v(L_{it})$

Consumption. A standard expansion gives:

$$\frac{u(C_t) - u(\bar{C})}{U_C \bar{C}} \overset{(2)}{=} y_t + \frac{1-\gamma}{2}\, y_t^2. \tag{B.15}$$

Labor disutility. Expanding $N_{it}\, v(L_{it}) = \frac{\alpha_i \zeta \bar{L}^{1+\varphi}}{1+\varphi} \exp\big(n_{it} + (1+\varphi) l_{it}\big)$ to second order, summing over i, dividing by $U_C \bar{C}$, and using identity (iii) to cancel the n_{it} and n_{it}^2 terms (exactly as in the static derivation):

$$\frac{\sum_i N_{it}\, v(L_{it}) - \sum_i \alpha_i\, v(\bar{L})}{U_C \bar{C}} \overset{(2)}{=} l_t^\alpha + \sum_i \alpha_i\, n_{it}\, h_{it} + \frac{1+\varphi}{2}\Big[(l_t^\alpha)^2 + \mathcal{M}_t\Big], \tag{B.16}$$

where $\mathcal{M}_t \equiv \sum_i \alpha_i\, h_{it}^2$.

Period return. Subtracting (B.16) from (B.15):

$$\frac{\big[u(C_t) - \sum_i N_{it}\, v(L_{it})\big] - \text{SS}}{U_C \bar{C}} \overset{(2)}{=} \underbrace{(y_t - l_t^\alpha)}_{\text{first-order}} + \underbrace{\frac{1-\gamma}{2}\, y_t^2 - \frac{1+\varphi}{2}\,(l_t^\alpha)^2}_{\text{aggregate quadratic}} - \underbrace{\frac{1+\varphi}{2}\,\mathcal{M}_t + \sum_i \alpha_i\, n_{it}\, h_{it}}_{\text{cross-sector}}. \tag{B.17}$$

We evaluate each group of terms in turn.

Aggregate quadratic. Since $l_t^\alpha \overset{(1)}{=} y_t - \bar{a}_t$ (from identity (i)), the aggregate quadratic in (B.17) is:

$$\frac{1-\gamma}{2}\, y_t^2 - \frac{1+\varphi}{2}\,(y_t - \bar{a}_t)^2 \;=\; -\frac{\gamma+\varphi}{2}\, y_t^2 + (1+\varphi)\, y_t\, \bar{a}_t - \frac{1+\varphi}{2}\, \bar{a}_t^2. \tag{B.18}$$

Substituting $y_t = y_t^{\text{nat}} + \tilde{y}_t$ with $y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi}\, \bar{a}_t$, the coefficient of $\tilde{y}_t$ is $-(\gamma+\varphi)\, y_t^{\text{nat}} + (1+\varphi)\, \bar{a}_t = 0$. Hence:

$$\text{(B.18)} \;=\; -\frac{\gamma+\varphi}{2}\, \tilde{y}_t^2 + (\text{t.i.p}), \tag{B.19}$$

with t.i.p denoting terms independent of policy.[8]

[8] The vanishing of the cross-term is the multi-sector counterpart of the standard result: the efficient-subsidy condition ensures that the first variation of $u - \sum Nv$ with respect to aggregate output vanishes at the natural level.

Cross-sector terms. Using identity (ii), the cross-sector contribution in (B.17) is, for each sector i (with the minus sign from the subtraction):

$$-\frac{1+\varphi}{2}\left(\chi_{it}+n_{it}\right)^2+n_{it}\left(\chi_{it}+n_{it}\right) \;=\; -\frac{1+\varphi}{2}\,\chi_{it}^2-\varphi\,\chi_{it}\,n_{it}-\frac{\varphi-1}{2}\,n_{it}^2. \tag{B.20}$$

Output–labor gap. The first-order term $y_t - l_t^{\alpha}$ in (B.17) has a nonzero second-order component. From the exact production identity (i) and the exogeneity of $\bar{a}_t$:

$$y_t - l_t^{\alpha} \;=\; \bar{a}_t+\sum_i \alpha_i\,n_{it}+\sum_i \alpha_i\,\xi_{it} \;\overset{(2)}{=}\; \text{t.i.p.}-\frac{1}{2}\sum_i \alpha_i\,n_{it}^2+\sum_i \alpha_i\,\xi_{it}, \tag{B.21}$$

using identity (iii). The Calvo price-dispersion recursion implies the present-value equivalence (assuming $\xi_{i,-1}=0$):

$$\sum_{t=0}^{\infty}\beta^t\sum_i \alpha_i\,\xi_{it} \;\overset{(2)}{=}\; -\sum_{t=0}^{\infty}\beta^t\sum_i \frac{\alpha_i\epsilon_i}{2\kappa_i}\,\pi_{it}^2. \tag{B.22}$$

Assembly. Substituting (B.19), (B.20), and (B.21) into (B.17), the period return relative to steady state is, in discounted-sum equivalence:

$$\begin{aligned}\frac{\left[u-\sum Nv\right]-\text{SS}}{U_C\bar{C}} \;\overset{(2)}{=}\; &\text{t.i.p.}-\frac{\gamma+\varphi}{2}\,\tilde{y}_t^2-\sum_i \frac{\alpha_i\epsilon_i}{2\kappa_i}\,\pi_{it}^2\\ &-\sum_i \alpha_i\left[\frac{1+\varphi}{2}\,\chi_{it}^2+\varphi\,\chi_{it}\,n_{it}+\frac{\varphi}{2}\,n_{it}^2\right],\end{aligned} \tag{B.23}$$

where the coefficient $\frac{\varphi}{2}$ on n_{it}^2 combines $\frac{\varphi-1}{2}$ from the cross-sector terms (B.20) and $\frac{1}{2}$ from the output–labor gap (B.21). The former reflects the convexity of the disutility of hours; the latter reflects the concavity of the Cobb–Douglas aggregator, which penalizes uneven employment even holding total labor input fixed.

Negating (B.23) and dropping terms independent of policy:

$$\mathcal{L}_t^{u-Nv} \;\overset{(2)}{=}\; \frac{\gamma+\varphi}{2}\,\tilde{y}_t^2+\sum_i \frac{\alpha_i\epsilon_i}{2\kappa_i}\,\pi_{it}^2+\sum_i \alpha_i\left[\frac{1+\varphi}{2}\,\chi_{it}^2+\varphi\,\chi_{it}\,n_{it}+\frac{\varphi}{2}\,n_{it}^2\right], \tag{B.24}$$

in discounted-sum equivalence.[9]

[9] Because the π_{it}^2 term arises from the present-value equivalence (B.22), equation (B.24) is the per-period term whose discounted sum equals the normalized welfare loss to second order.

B.2.2 Mobility Cost Telescopes into a Startup Term

Step 1 expanded $u(C_t) - \sum_i N_{it}\, v(L_{it})$. The full per-period utility also includes the mobility cost $-\Psi_t$. We now construct a sector-specific net continuation value that strips out future flow utility, yielding a boundary term that enters the Ramsey Lagrangian only through the initial values ignored in the timeless perspective.

Sector-specific net continuation value. For each sector i and date t, define the flow-utility continuation

$$F_{it} \;\equiv\; U_{it}^{\text{flow}} + \beta \sum_{j \in \mathcal{N}} \mu_t^{ij}\, \mathbb{E}_t\, F_{jt+1}, \tag{B.25}$$

and the net continuation value

$$\check{V}_{it} \;\equiv\; V_{it} - F_{it}. \tag{B.26}$$

Intuitively, F_{it} captures the expected discounted flow utility a worker in sector i would earn along the realized transition path, and $\check{V}_{it}$ isolates the remaining component of continuation value—the part attributable to the option value of reallocation net of mobility costs.

Lemma B.1. *Let $\check{\mathcal{B}}_t \equiv \sum_i N_{it}\, \check{V}_{it}$. Then:*

$$-\Psi_t \;=\; \check{\mathcal{B}}_t - \beta\, \mathbb{E}_t\, \check{\mathcal{B}}_{t+1}. \tag{B.27}$$

Consequently, if $\lim_{T\to\infty} \beta^T\, \mathbb{E}_0 |\check{\mathcal{B}}_T| = 0$, then

$$\mathcal{W}_0 \;=\; \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[u(C_t) - \sum_i N_{it}\, v(L_{it}) \Big] \;+\; \check{\mathcal{B}}_0. \tag{B.28}$$

Proof. The Bellman equation (13), rewritten using the logit probabilities (12) and the reallocation cost (11), takes the envelope form:

$$V_{it} \;=\; U_{it}^{\text{flow}} + \beta \sum_j \mu_t^{ij}\, \mathbb{E}_t\, V_{jt+1} - \psi_i\Big(\{\mu_t^{ij}\}_j \Big). \tag{B.29}$$

Subtracting the F_{it} recursion (B.25) from (B.29), the flow-utility terms cancel:

$$\check{V}_{it} \;=\; \beta \sum_j \mu_t^{ij}\, \mathbb{E}_t\, \check{V}_{jt+1} - \psi_i\Big(\{\mu_t^{ij}\}_j \Big). \tag{B.30}$$

Multiplying by N_{it}, summing over i, and using $N_{jt+1} = \sum_i N_{it}\, \mu_t^{ij}$:

$$\sum_i N_{it}\, \check{V}_{it} \;=\; \beta\, \mathbb{E}_t \sum_j N_{jt+1}\, \check{V}_{jt+1} - \sum_i N_{it}\, \psi_i\Big(\{\mu_t^{ij}\}_j\Big)\,,$$

which is $\check{\mathcal{B}}_t = \beta\, \mathbb{E}_t\, \check{\mathcal{B}}_{t+1} - \Psi_t$, giving (B.27). Multiplying by β^t, summing, and applying the transversality condition telescopes the mobility cost into $-\check{\mathcal{B}}_0$. Substituting into $\mathcal{W}_0 = \mathbb{E}_0 \sum_{t\geq 0} \beta^t [u - \sum Nv - \Psi_t]$ yields (B.28). □

Equation (B.28) is exact: the entire contribution of the mobility block to lifetime welfare is captured by $\check{\mathcal{B}}_0 = \sum_i N_{i0}\, \check{V}_{i0}$. The key property of this construction is that $\check{\mathcal{B}}_0$ is a *pure startup term*: it depends on the future allocation only through the initial net continuation values $\{\check{V}_{i0}\}$, which are themselves determined by the recursion (B.30). In the Ramsey problem with $\{\check{V}_{it}\}$ treated as auxiliary variables subject to this recursion, $\check{\mathcal{B}}_0$ affects only the date-0 first-order conditions—it is the source of the well-known tension between the optimal long-run policy rule and the incentive to deviate at the initial date.

Under the timeless perspective (Woodford, 2003), the planner commits to a policy rule that would have been optimal had it been adopted far in the past. This amounts to requiring that the same targeting criterion hold at every date, including $t = 0$, rather than exploiting the special initial conditions. $\check{\mathcal{B}}_0$ affects only the exceptional startup conditions of the date-0 Ramsey problem. Under the timeless perspective, these startup conditions are replaced by the same self-consistent targeting conditions that apply for $t \geq 1$, and the per-period loss for the Ramsey problem under the timeless perspective is therefore $\mathcal{L}_t^{u-Nv}$ from (B.24).

Combining Steps 1 and 2 and dropping terms independent of policy, the per-period loss for the Ramsey problem is:

$$\mathcal{L}_t \;\overset{(2)}{=}\; \frac{\gamma+\varphi}{2}\, \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i}\, \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1+\varphi}{2}\, \chi_{it}^2 + \varphi\, \chi_{it}\, n_{it} + \frac{\varphi}{2}\, n_{it}^2\right], \tag{B.31}$$

in discounted-sum equivalence, which is equation (68). □

Remark 3 (Consistency with the static loss)**.** *In the static limit, substituting $n_{it} = -\frac{\phi}{1+\phi}\, \chi_{it}$ into the cross-sector terms of (B.31) yields a coefficient on χ_{it}^2 that falls short of the static loss coefficient $\frac{\sigma}{2}$ (Proposition 1) by exactly $\frac{\varphi\phi}{2(1+\phi)^2}$. This difference equals the second-order contribution of $-(\check{\mathcal{B}}_0 - \check{\mathcal{B}}_0^{nat})/(U_C\bar{C})$ to the loss. In the static limit $\check{V}_{it} = -\psi_i(\{\bar{\mu}^{ij}\})$, so $-(\check{\mathcal{B}}_0 - \check{\mathcal{B}}_0^{nat})/(U_C\bar{C})$ reduces to the per-period mobility cost $\frac{\varphi\phi}{2}\, \mathcal{M}_t$ derived in equation (B.6). In the static economy there is no intertemporal reallocation block, so this boundary contribution enters the per-period loss directly, recovering the σ coefficient. In the dynamic economy, the same welfare cost is absorbed by $\check{\mathcal{B}}_0$ and internalized through the reallocation-block multipliers $(\boldsymbol{\psi}_t, \boldsymbol{\xi}_t)$ in the Ramsey Lagrangian.*

C Additional Proofs of Propositions and Lemmas

C.1 Section 4

Proof of Lemma 4. We prove each statement sequentially.

Natural output and sectoral hours. Under flexible prices, the efficient subsidy eliminates markups and the intratemporal first-order condition (32) gives, in log-deviations:

$$(1-\gamma)\, y_t^{\text{nat}} = n_{it} + (1+\varphi)\, l_{it}^{\text{nat}}, \qquad \forall\, i. \tag{C.1}$$

Since $\sum_i \alpha_i n_{it} = 0$, weighting by α_i and summing yields $y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \sum_i \alpha_i a_{it}$, identical to the static case and independent of $\mathbf{n}_t$. Solving (C.1) for sectoral hours gives $l_{it}^{\text{nat}} = \bar{l}_t^{\text{nat}} - \frac{1}{1+\varphi} n_{it}$, where $\bar{l}_t^{\text{nat}} \equiv \frac{1-\gamma}{1+\varphi} y_t^{\text{nat}}$.

Natural relative prices. Goods market clearing gives $\hat{p}_{it}^{\text{nat}} = y_t^{\text{nat}} - a_{it} - l_{it}^{\text{nat}} - n_{it}$. Substituting (48):

$$\hat{p}_{it}^{\text{nat}} = \frac{\gamma+\varphi}{1+\varphi}\, y_t^{\text{nat}} - a_{it} - \frac{\varphi}{1+\varphi}\, n_{it} = \left(\sum_j \alpha_j a_{jt} - a_{it} \right) - \frac{\varphi}{1+\varphi}\, n_{it},$$

where the second equality uses $\frac{\gamma+\varphi}{1+\varphi}\, y_t^{\text{nat}} = \sum_j \alpha_j a_{jt}$.

Natural real rate. Since y_t^{nat} is independent of $\mathbf{n}_t$, the Euler equation (9) gives $r_t^{\text{nat}} = -\ln\beta + \gamma\, \mathbb{E}_t(y_{t+1}^{\text{nat}} - y_t^{\text{nat}})$, identical to the static case.

Marginal cost decomposition. The log-linearized marginal cost (22) is $mc_{it} - p_{it} = (\gamma+\varphi) y_t - (1+\varphi)(\hat{p}_{it} + a_{it}) - \varphi\, n_{it}$. Defining $\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}$ and using $y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \sum_i \alpha_i a_{it}$, the natural-level marginal cost satisfies $mc_{it}^{\text{nat}} = p_{it}^{\text{nat}}$ (by the efficient subsidy). Taking the gap $\tilde{y}_t = y_t - y_t^{\text{nat}}$ and noting that $\hat{p}_{it} + a_{it} = \chi_{it} - \hat{a}_{it} + a_{it} = \chi_{it} + \sum_j \alpha_j a_{jt}$:

$$\begin{aligned} mc_{it} - p_{it} &= (\gamma+\varphi)\, \tilde{y}_t + (\gamma+\varphi)\, y_t^{\text{nat}} - (1+\varphi)\left(\chi_{it} + \textstyle\sum_j \alpha_j a_{jt} \right) - \varphi\, n_{it} \\ &= (\gamma+\varphi)\, \tilde{y}_t - (1+\varphi)\, \chi_{it} - \varphi\, n_{it}, \end{aligned}$$

where the second line uses $(\gamma+\varphi)\, y_t^{\text{nat}} = (1+\varphi) \sum_j \alpha_j a_{jt}$.

Natural reallocation dynamics. We additionally characterize the law of motion for $\mathbf{n}_t$ under flexible prices. The demeaned flow utility at the natural allocation is:

$$\hat{u}_{it}^{\text{nat}} = l_{it}^{\text{nat}} - \bar{l}_t^{\text{nat}} = -\frac{1}{1+\varphi}\, n_{it},$$

using (48) and the fact that the aggregate component cancels upon demeaning. Workers in overstaffed sectors receive lower flow utility and hence have incentives to leave.

Substituting $\hat{\mathbf{u}}_t^{\text{nat}} = -\frac{1}{1+\varphi}\,\mathbf{n}_t^{\text{nat}}$ into the value recursion (26) and iterating forward, the continuation value under flexible prices satisfies:

$$\hat{\mathbf{v}}_t^{\text{nat}} = -\frac{1}{1+\varphi}\sum_{s=0}^{\infty}(\beta\bar{\mu})^s\,\mathbb{E}_t[\mathbf{n}_{t+s}^{\text{nat}}]. \tag{C.2}$$

Suppose the natural law of motion takes the linear form $\mathbf{n}_{t+1}^{\text{nat}} = F\,\mathbf{n}_t^{\text{nat}}$ for some operator F to be determined. Then $\mathbb{E}_t[\mathbf{n}_{t+s}^{\text{nat}}] = F^s\,\mathbf{n}_t^{\text{nat}}$ and:

$$\hat{\mathbf{v}}_t^{\text{nat}} = -\frac{1}{1+\varphi}(I - \beta\bar{\mu}\,F)^{-1}\,\mathbf{n}_t^{\text{nat}}.$$

Substituting into the employment evolution (25):

$$\mathbf{n}_{t+1}^{\text{nat}} = \bar{\mu}\,\mathbf{n}_t^{\text{nat}} - \frac{\phi\beta(I-\bar{\mu}^2)}{1+\varphi}(I-\beta\bar{\mu}\,F)^{-1}F\,\mathbf{n}_t^{\text{nat}} \equiv F\,\mathbf{n}_t^{\text{nat}}. \tag{C.3}$$

which is a fixed-point characterization of F. To verify that F implies mean-reversion, diagonalize $\bar{\mu}$. Under detailed balance (Assumption 1), $\bar{\mu}$ is similar to a symmetric matrix and hence has real eigenvalues $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_N > -1$ (cannot equal -1 by Perron-Frobenius Theorem). Along the k-th eigenvector (for $k \geq 2$), the fixed-point condition (C.3) reduces to the scalar equation:

$$f_k = \lambda_k - \frac{\phi\beta(1-\lambda_k^2)f_k}{(1+\varphi)(1-\beta\lambda_k f_k)}. \tag{C.4}$$

where f_k is the eigenvalue of F along component k. Rearranging, f_k solves the quadratic:

$$\beta\lambda_k\,f_k^2 - \left[1+\beta\lambda_k^2+\frac{\phi\beta(1-\lambda_k^2)}{1+\varphi}\right]f_k + \lambda_k = 0. \tag{C.5}$$

The product of roots is $\lambda_k/(\beta\lambda_k) = 1/\beta > 1$, so one root is inside and one outside the unit circle. The stable root $f_k \in (0, \lambda_k)$ gives the natural rate of employment mean-reversion along component k. That $f_k < \lambda_k$ confirms that directed reallocation (governed by ϕ) accelerates

convergence beyond what passive baseline transitions (governed by $\bar{\mu}$) alone would achieve. $\square$

Proof of Lemma 6. The proof proceeds in three steps: symmetrization, spectral properties, and decoupling.

Step 1: Symmetrization. Detailed balance (16) states $\alpha_i \bar{\mu}^{ij} = \alpha_j \bar{\mu}^{ji}$ for all i, j, which in matrix form is $A\bar{\mu} = \bar{\mu}^\top A$ where $A = \mathrm{diag}(\alpha_i)$. Define $Q \equiv A^{1/2} \bar{\mu}\, A^{-1/2}$. Then:

$$Q^\top = A^{-1/2} \bar{\mu}^\top A^{1/2} = A^{-1/2}(A\bar{\mu}\, A^{-1}) A^{1/2} = A^{1/2} \bar{\mu}\, A^{-1/2} = Q,$$

where the second equality uses $\bar{\mu}^\top = A\bar{\mu}\, A^{-1}$ (from $A\bar{\mu} = \bar{\mu}^\top A$). So Q is real symmetric.

Step 2: Spectral properties. Since Q is real symmetric, it has real eigenvalues $\rho_1 \geq \rho_2 \geq \cdots \geq \rho_N$ and orthonormal eigenvectors $\{s_k\}_{k=1}^N$. We establish the claimed ordering.

Dominant eigenvalue. Since $\bar{\mu}$ is a row-stochastic matrix, $\bar{\mu}\,\mathbf{1} = \mathbf{1}$. Therefore $Q(A^{1/2}\mathbf{1}) = A^{1/2}\bar{\mu}\,\mathbf{1} = A^{1/2}\mathbf{1}$, so $s_1 = A^{1/2}\mathbf{1}/\|A^{1/2}\mathbf{1}\|$ is an eigenvector with eigenvalue $\rho_1 = 1$.

Strict bounds. Under the logit specification (12), $\bar{\mu}^{ij} > 0$ for all i, j (the exponential is always positive). Therefore $\bar{\mu}$ is a strictly positive stochastic matrix. By the Perron–Frobenius theorem, $\rho_1 = 1$ is a simple eigenvalue and all other eigenvalues satisfy $|\rho_k| < 1$ for $k \geq 2$, which gives $1 = \rho_1 > \rho_2 \geq \cdots \geq \rho_N > -1$.

Stochastic bounds. Since Q is similar to the stochastic matrix $\bar{\mu}$ (via the conjugation $A^{\pm 1/2}$), they share the same eigenvalues. All eigenvalues of a stochastic matrix have modulus at most one, confirming $|\rho_k| \leq 1$.

Step 3: Decoupling the relative block. Define the coordinate transformation $x^*_{kt} \equiv s_k^\top A^{1/2} \hat{\mathbf{x}}_t$ for each variable $x \in \{\chi, n, \hat{v}, \hat{a}\}$. Since $\hat{\mathbf{x}}_t = M\mathbf{x}_t$ where $M = I - \mathbf{1}\boldsymbol{\alpha}^\top$, we have $\boldsymbol{\alpha}^T \hat{x}_t = 0$. Then

$$x^*_{1t} = s_1^T A^{1/2} \hat{x}_t \propto (A^{1/2}\mathbf{1})^T A^{1/2} \hat{\mathbf{x}}_t = 1^T A \hat{\mathbf{x}}_t = \alpha^T \hat{\mathbf{x}}_t = 0$$

So only components $k \geq 2$ are nontrivial.

We now show that each equation in the relative block (53)–(56) decouples. The key identity is that whenever $\bar{\mu}$ or $\bar{\mu}^2$ appears, the transformation diagonalizes it:

$$s_k^\top A^{1/2} \bar{\mu}\, \hat{\mathbf{x}}_t = s_k^\top A^{1/2} \bar{\mu}\, A^{-1/2} A^{1/2} \hat{\mathbf{x}}_t = s_k^\top Q\, A^{1/2} \hat{\mathbf{x}}_t = \rho_k\, x^*_{kt},$$

using $Q\, s_k = \rho_k\, s_k$ and $s_k^\top s_j = \delta_{kj}$. Similarly, $s_k^\top A^{1/2} \bar{\mu}^2 \hat{\mathbf{x}}_t = \rho_k^2\, x^*_{kt}$ and $s_k^\top A^{1/2}(I - \bar{\mu}^2)\hat{\mathbf{x}}_t = (1 - \rho_k^2)\, x^*_{kt}$.

Applying $s_k^\top A^{1/2}$ to each equation in the relative block (with constant κ):

Relative NKPC (53): $s_k^\top A^{1/2}\hat{\boldsymbol{\pi}}_t = -\kappa[(1+\varphi)\,\chi_{kt}^* + \varphi\, n_{kt}^*] + \beta\,\mathbb{E}_t[\pi_{kt+1}^*]$, where we used $\kappa_i = \kappa$ to factor κ out of the projection.

Relative price evolution (54): $\chi_{kt}^* = \chi_{k,t-1}^* + \pi_{kt}^* + \Delta\hat{a}_{kt}^*$, which is immediate since M and the identity commute with the projection.

Employment evolution (55): $n_{k,t+1}^* = \rho_k\, n_{kt}^* + \phi\beta(1-\rho_k^2)\,\mathbb{E}_t[\hat{v}_{k,t+1}^*]$, using the diagonalization identities above.

Value recursion (56): $\hat{v}_{kt}^* = -\chi_{kt}^* - n_{kt}^* + \beta\rho_k\,\mathbb{E}_t[\hat{v}_{k,t+1}^*]$.

Each projected equation for component k involves only (χ_{kt}^*, n_{kt}^*, $\hat{v}_{kt}^*$) and the exogenous forcing $\hat{a}_{kt}^*$, with no coupling to any $k' \neq k$. The constant-κ assumption is used only in the NKPC projection: with heterogeneous κ_i, the term $s_k^\top A^{1/2}\,\mathrm{diag}(\kappa_i)\hat{\boldsymbol{\chi}}_t$ would involve off-diagonal entries $s_k^\top A^{1/2} K A^{-1/2} s_{k'}$ for $k' \neq k$, coupling the components. □

Proof of Proposition 4. We evaluate the persistence, impact, and cumulative inflation in turn.

Persistence. Let $f(\lambda;\Gamma) = \beta\lambda^2 - \Gamma\lambda + 1$. The stable root $\lambda \in (0,1)$ decreases strictly in Γ. Comparing (60) and (61), the segmented coefficient $\Gamma_{\text{seg}} = 1 + \beta + \kappa(1+\varphi)$ is strictly larger than the integrated coefficient $\Gamma_{\text{int}} = 1 + \beta + \kappa$ because $\kappa\varphi > 0$. Therefore, $\lambda_{\text{seg}} < \lambda_{\text{int}}$, proving that segmented markets decay faster.

Impact. The characteristic equation (60) implies $\lambda_{\text{seg}} = (\Gamma_{\text{seg}} - \beta\lambda_{\text{seg}})^{-1}$. Therefore, the initial distortion under segmented markets is $\chi_{j0,\text{seg}}^* = \hat{a}/(\Gamma_{\text{seg}} - \beta\lambda_{\text{seg}})$. Comparing this to the integrated initial condition in (62):

$$\chi_{j0,\text{int}}^* = \frac{1}{\Gamma_{\text{seg}} - \beta\lambda_{\text{int}}}\,\hat{a}$$

Because $\lambda_{\text{seg}} < \lambda_{\text{int}}$, the denominator for the integrated market is smaller, meaning $\chi_{j0,\text{seg}}^* < \chi_{j0,\text{int}}^*$. Since impact inflation is $\hat{\pi}_{j0}^* = \chi_{j0}^* - \hat{a}$, its magnitude is $|\hat{\pi}_{j0}^*| = \hat{a} - \chi_{j0}^*$. The smaller initial distortion under segmented markets thus yields a strictly larger initial inflation impact: $|\hat{\pi}_{j0,\text{seg}}^*| > |\hat{\pi}_{j0,\text{int}}^*|$.

Cumulative. By definition, relative inflation is the change in the relative price level, $\hat{\pi}_{jt}^* = \hat{p}_{jt}^* - \hat{p}_{jt-1}^*$. The cumulative sum telescopes: $\sum_{t=0}^{\infty}\hat{\pi}_{jt}^* = \lim_{t\to\infty}\hat{p}_{jt}^* - \hat{p}_{j,-1}^*$. Starting from steady state ($\hat{p}_{j,-1}^* = 0$), prices must eventually clear the market to eliminate the distortion ($\lim_{t\to\infty}\chi_{jt}^* = 0$). Since $\chi_{jt}^* = \hat{p}_{jt}^* + \hat{a}$, the relative price level must converge to $\lim_{t\to\infty}\hat{p}_{jt}^* = -\hat{a}$, yielding $\sum_{t=0}^{\infty}\hat{\pi}_{jt}^* = -\hat{a}$ regardless of ϕ. □

Proof of Proposition 5. The proof proceeds in four steps: derivation of the characteristic polynomial, root counting, and verification of the two limiting cases.

Step 1: Characteristic polynomial. After shocks have died out ($\hat{a}^*_{kt} = 0$), conjecture $\chi^*_{kt} = \lambda^t C$, $n^*_{kt} = \lambda^t N$, $\hat{v}^*_{kt} = \lambda^t V$ for constants C, N, V. Substituting into the component-k system (57)–(59):

$$\left[\beta\lambda^2 - (\kappa(1+\varphi) + \beta + 1)\lambda + 1\right] C = \kappa\varphi\lambda\, N, \tag{C.6}$$

$$(\lambda - \rho_k)\, N = \phi\beta(1-\rho_k^2)\lambda\, V, \tag{C.7}$$

$$V = -(C+N) + \beta\rho_k\lambda\, V. \tag{C.8}$$

Solving (C.8) for V:

$$V = -\frac{C+N}{1-\beta\rho_k\lambda}. \tag{C.9}$$

Substituting (C.9) into (C.7):

$$(\lambda - \rho_k)\, N = -\frac{\phi\beta(1-\rho_k^2)\lambda}{1-\beta\rho_k\lambda}\,(C+N).$$

Multiplying through by $(1-\beta\rho_k\lambda)$ and collecting:

$$\left[(\lambda-\rho_k)(1-\beta\rho_k\lambda) + \phi\beta(1-\rho_k^2)\lambda\right] N = -\phi\beta(1-\rho_k^2)\lambda\, C. \tag{C.10}$$

Using (C.10) to express N in terms of C and substituting into (C.6) yields, after cancelling $C \neq 0$:

$$\begin{aligned}\left[(\lambda-\rho_k)(1-\beta\rho_k\lambda) + \phi\beta(1-\rho_k^2)\lambda\right]&\left[\beta\lambda^2 - (\kappa(1+\varphi)+\beta+1)\lambda + 1\right]\\ &+ \kappa\varphi\phi\beta(1-\rho_k^2)\lambda^2 = 0,\end{aligned}$$

which is (63). Expanding, the first bracket is quadratic in λ (with leading term $-\beta\rho_k\lambda^2$) and the second bracket is quadratic (with leading term $\beta\lambda^2$), so the product is degree four. The final additive term is degree two, preserving the degree. Hence (63) is a degree-4 polynomial $p(\lambda) = 0$.

Step 2: Root count. The system has three variables $(\chi^*_{kt}, n^*_{kt}, \hat{v}^*_{kt})$ and one lag from the NKPC (through $\chi^*_{k,t-1}$), giving a total state dimension of four. By the Blanchard–Kahn condition, a unique bounded solution requires exactly two stable roots ($|\lambda| < 1$) and two unstable roots ($|\lambda| > 1$).[10]

[10] The degree-4 polynomial (63) can be verified numerically to have exactly two roots inside the unit circle across the economically relevant parameter space.

Step 3: Limiting case $\phi \to 0$. Setting $\phi = 0$ in (63):

$$(\lambda - \rho_k)(1 - \beta\rho_k\lambda)\left[\beta\lambda^2 - (\kappa(1+\varphi) + \beta + 1)\lambda + 1\right] = 0.$$

This factors into four explicit roots. The first factor gives $\lambda = \rho_k$ (stable, since $|\rho_k| < 1$). The second gives $\lambda = 1/(\beta\rho_k)$ (unstable, since $|\rho_k| < 1$). The quadratic is the segmented-market characteristic equation (60), whose roots have product $1/\beta > 1$: one stable root $\lambda_{\text{seg}} \in (0, 1)$ and one unstable root $1/(\beta\lambda_{\text{seg}}) > 1$. The two stable roots are therefore $\{\lambda_{\text{seg}}, \rho_k\}$, and the spectral radius of the stable block is $\lambda^*(0) = \max\{\lambda_{\text{seg}}, |\rho_k|\}$.

Step 4: Limiting case $\phi \to \infty$. Divide (63) by ϕ and take $\phi \to \infty$. All terms not proportional to ϕ vanish, leaving:

$$\beta(1 - \rho_k^2)\lambda\left[\beta\lambda^2 - (\kappa(1+\varphi) + \beta + 1)\lambda + 1\right] + \kappa\varphi\beta(1 - \rho_k^2)\lambda^2 = 0.$$

Since $\beta(1 - \rho_k^2)\lambda \neq 0$ for $\lambda \neq 0$ and $|\rho_k| < 1$, divide through:

$$\beta\lambda^2 - (\kappa(1+\varphi) + \beta + 1)\lambda + 1 + \kappa\varphi\lambda = 0,$$

which simplifies to

$$\beta\lambda^2 - (\kappa + \beta + 1)\lambda + 1 = 0.$$

This is the integrated-market characteristic equation (61), whose stable root is λ_{int}. The fourth root at $\lambda = 0$ (from the factor λ extracted above) confirms that one degree of freedom collapses as $\phi \to \infty$: employment fully tracks relative prices ($n_{kt}^* = -\chi_{kt}^*$), eliminating one independent state variable.

Continuity. The roots of (63) depend continuously on ϕ (as roots of a polynomial with coefficients that are continuous in ϕ). Since the dominant stable root $\lambda^*(\phi)$ remains bounded away from the unit circle for all finite $\phi > 0$ (by Blanchard–Kahn), $\lambda^*(\phi)$ is a continuous function of ϕ on $[0, \infty)$, interpolating between the two limiting values. □

Proof of Proposition 6. We proceed in steps.

Part (i): Effective slope. Write $K = \text{diag}(\kappa_i) = \kappa I + \varepsilon\ \text{diag}(\Delta\kappa_i)$. Under constant κ, the NKPC projection onto component k (Lemma 6) involves $s_k^\top\ \text{diag}(\kappa_i)\, s_{k'} = \kappa\, \delta_{kk'}$. Under the perturbation,

$$s_k^\top\ \text{diag}(\kappa_i)\, s_{k'} = \kappa\, \delta_{kk'} + \varepsilon \sum_i \Delta\kappa_i\, s_{k,i}\, s_{k',i}.$$

Using $s_{k,i} = \alpha_i^{1/2} v_{k,i}$, the diagonal entry is $\kappa + \varepsilon\,\Delta\kappa_k^{\text{eff}}$. Off-diagonal entries are $O(\varepsilon)$ but affect the eigenvalue only at $O(\varepsilon^2)$ by standard non-degenerate perturbation theory. To first order, the component-k polynomial is therefore obtained by replacing κ with $\kappa + \varepsilon\,\Delta\kappa_k^{\text{eff}}$ in (63).

Part (ii): Persistence shift. The dominant stable root λ_k^0 satisfies $p(\lambda_k^0;\, \kappa,\, \rho_k) = 0$. By Part (i), the perturbed polynomial is $p(\lambda;\, \kappa + \varepsilon\,\Delta\kappa_k^{\text{eff}},\, \rho_k) + O(\varepsilon^2) = 0$. The implicit function theorem at the simple root λ_k^0 gives (65) with $\Gamma_k = -\partial_\kappa p / \partial_\lambda p$ evaluated at $(\lambda_k^0, \kappa, \rho_k)$.

Formula and Sign of Γ_k. Decompose (63) as $p = R \cdot S + T$ where

$$\begin{aligned} R(\lambda) &\equiv (\lambda - \rho_k)(1 - \beta\rho_k\lambda) + \phi\beta(1 - \rho_k^2)\lambda, \\ S(\lambda;\kappa) &\equiv \beta\lambda^2 - [\kappa(1+\varphi) + \beta + 1]\lambda + 1, \\ T(\lambda;\kappa) &\equiv \kappa\varphi\phi\beta(1 - \rho_k^2)\lambda^2. \end{aligned}$$

Since T is linear in κ, $\partial_\kappa T = T/\kappa$. Using $p(\lambda_k^0) = 0$, i.e. $T(\lambda_k^0) = -R(\lambda_k^0)S(\lambda_k^0)$, the κ-derivative simplifies to

$$\partial_\kappa\, p\big|_{\lambda_k^0} = -(1+\varphi)\lambda_k^0\, R(\lambda_k^0) + \tfrac{1}{\kappa}\, T(\lambda_k^0) = -\frac{R(\lambda_k^0)}{\kappa}\Big[\kappa(1+\varphi)\lambda_k^0 + S(\lambda_k^0)\Big].$$

Substituting the definition of S collapses the bracket:

$$\kappa(1+\varphi)\lambda_k^0 + S(\lambda_k^0) = \beta(\lambda_k^0)^2 - (\beta+1)\lambda_k^0 + 1 = (\lambda_k^0 - 1)(\beta\lambda_k^0 - 1) > 0,$$

where the inequality uses $\lambda_k^0 < 1$ and $\beta < 1$. For $\rho_k \in (0,1)$, $T(\lambda_k^0) > 0$ and $RS = -T$ imply $R(\lambda_k^0) > 0$ and $S(\lambda_k^0) < 0$. Together, $\partial_\kappa p|_{\lambda_k^0} < 0$. Finally, the degree-4 polynomial satisfies $p(0) > 0$ and $p(1) < 0$, so it crosses zero from above at the dominant stable root: $\partial_\lambda p|_{\lambda_k^0} < 0$. Therefore

$$\Gamma_k = -\frac{\partial_\kappa\, p}{\partial_\lambda\, p}\bigg|_{\lambda_k^0} < 0:$$

a higher effective slope strictly accelerates convergence. □